The Salinas River Watershed

A Preliminary Sustainable Watershed Management Plan

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Executive Summary

The Salinas River Watershed is a nationally significant agricultural region producing nearly half the nation's lettuce, celery, broccoli, and spinach, with over $5 billion in annual agricultural output. This diverse and productive system supports intensive farming operations, urban communities including Salinas, and ecologically sensitive habitats such as Elkhorn Slough and the Monterey Bay National Marine Sanctuary. However, the watershed faces critical sustainability challenges that threaten the balance between agricultural productivity, ecological health, and long-term water availability.

Primary Challenges

The watershed's primary challenges stem from the intersection of intensive land use and limited water resources.

1. **Groundwater overdraft** represents the most pressing concern, with the primary aquifer systems supplying municipal and industrial water critically overdrafted and vulnerable to seawater intrusion due to the basin's coastal proximity. Water use is dominated by agricultural irrigation, which accounts for 44% of total withdrawals and over 90% of groundwater pumping.
2. **Water quality degradation** driven by non-point source pollution compounds these supply challenges. Agricultural activities generate the majority of pollutant loading across the watershed such as sediment, biochemical oxygen demand, nitrogen and phosphorus. These nutrient and organic matter inputs pose significant eutrophication risks, particularly for downstream systems where dilution is limited.
3. **Monitoring infrastructure inadequacies** further complicate management efforts. Significant monitoring gaps exist in the southeastern areas around Gonzales and Soledad, limiting the ability to trace pollutant sources, assess upstream conditions, and attribute impacts.

Key Objectives

The Salinas River Watershed Management Plan addresses these interconnected challenges through coordinated strategies that align land use planning, water supply sustainability, flood management, and water quality protection under relevant regulatory frameworks including SGMA (Sustainable Groundwater Management Act) and TMDL requirements. This plan seeks to maintain the region's role as a national agricultural leader while ensuring long-term ecological and hydrological sustainability. Success depends on coordinated implementation of strategies that address both human and environmental needs:

1. **Managing groundwater extraction to prevent further overdraft and seawater intrusion**

a. Groundwater sustainability is managed through the Salinas Valley Basin Groundwater Sustainability Agency and coordinated Groundwater Sustainability Plans under SGMA.
b. Gonzales has established a policy cap near 2010 extraction levels for buildout, requiring conservation, recycling, and reuse rather than increased pumping.
c. County policy links discretionary development approvals to proof of long-term sustainable water supplies in both quality and quantity, ensuring that growth does not exacerbate overdraft conditions.

2. **Controlling non-point source pollution through comprehensive agricultural BMPs**
    a. Priority implementation areas include the intensively farmed central valley corridor where cropland and pasture dominate, and urbanized areas around Salinas with high impervious cover. Recommended BMPs align closely with ongoing regional initiatives including the Castroville to the Coast Project, the Old Salinas River Project, Espinosa Lake floodplain expansion, and the Salinas River Stream Maintenance Program, all emphasizing habitat restoration, flood resilience, and water quality improvement.
    b. Based on pollutant loads, land-use distribution, and source attribution, priority BMPs include riparian forest and grass buffers, improved nutrient management, conservation tillage, rotational grazing, streambank stabilization, and urban practices such as bioretention and infiltration devices. These measures directly target the high BOD, nitrogen, and sediment loads generated from cropland, pastureland, and urban sources.
    c. Successful implementation requires targeted BMP deployment, preservation of riparian corridors, focused investigations at pollutant hotspots, and expanded monitoring to address current gaps—particularly in the upper watershed where source attribution remains limited.
3. **Aligning urban growth with sustainable water supplies**
    a. Land use policies for Salinas, Soledad and Gonzales follow a city-centered growth approach that channels development to urban areas while protecting agricultural landscapes. These frameworks create quantifiable demand scenarios for Urban Water Management Plans and align growth with water infrastructure capacity.
    b. Flood control policies restrict development in the 100-year floodplain and maintain channel capacity in the Salinas and Arroyo Seco rivers for flood conveyance, groundwater recharge, and riparian habitat functions. Local ordinances require low-impact development and best management practices (BMPs) to retain runoff from frequent storms and detain larger events, with open space corridors serving both protection and recreation objectives.
4. **Expanding monitoring infrastructure to support adaptive management.**
    a. Expand water quality monitoring coverage through the CCAMP and Cooperative Monitoring Program (CMP) networks to address spatial gaps in the upper watershed, enabling better tracking of nitrate, ammonia, orthophosphate, dissolved oxygen, pesticides, and sediment toxicity.

b. Establish additional gaging stations in the upper watershed, particularly in the SE areas around Gonzales and Soledad, to complement the existing Spreckels and Chualar stations and improve source attribution for pollutant loading.
c. Integrate groundwater and surface water monitoring to better understand the connection between the 180 and 400-foot aquifer systems and surface water quality, supporting coordinated SGMA implementation.

Through integrated analysis of land use, hydrology, geology, climate, and water quality, the Plan provides a framework to guide decisions. These spatially targeted strategies complement existing efforts and provide a scalable framework for protecting beneficial uses that encompass municipal and agricultural supply, industrial uses, recreation, warm freshwater habitats, spawning and wildlife habitat, and rare species protection within this essential California watershed.

Section 1: Watershed Characterization

1.1 Introduction

The Salinas River Watershed encompasses a diverse landscape shaped by geology, climate, and human land use that collectively define its hydrologic and ecological character. Bounded by the Santa Lucia and Gabilan mountain ranges, the watershed drains approximately 4,600 square miles across Monterey and San Luis Obispo Counties before emptying into Monterey Bay. A Mediterranean climate with seasonal precipitation concentrated between November and March drives streamflow patterns and groundwater recharge dynamics across the basin. The valley floor, transformed by intensive agricultural development into the nation's "Salad Bowl," overlies a complex layered aquifer system that has become increasingly stressed by chronic groundwater overdraft and seawater intrusion. Surrounding this agricultural core, diverse ecosystems—from high-elevation coniferous forests to coastal wetlands and estuaries—support regionally significant biological communities and provide critical habitat for threatened and endangered species. This section examines the physical, hydrologic, and biological characteristics of the Salinas Watershed that together establish both the opportunities and constraints for sustainable water management.

1.2 Location of the Watershed

Understanding the location and extent of the Salinas River Watershed within California's broader hydrologic and administrative landscape is fundamental to effective management planning. Displayed in **Figure 1.2.1**, The watershed's position on the Central Coast places it within a region of significant agricultural, economic, and ecological importance. Recognizing where the Salinas Watershed fits within the state's geography—relative to neighboring watersheds, county boundaries, and coastal ecosystems—provides essential context for understanding water availability, regulatory frameworks, and stakeholder interests that shape management decisions.

Figure 1.2.1: Salinas River Watershed Location within California. The Salinas River Watershed is located on California's Central Coast in Monterey and San Luis Obispo Counties. The watershed drains approximately 4,600 square miles and is bounded by the Santa Lucia Range to the west and the Gabilan Range to the east, with the Salinas River flowing northwestward approximately 90 miles to Monterey Bay. The valley floor encompasses the primary agricultural production area and municipal water service districts, including the cities of Salinas, Gonzales, and Soledad. The Elkhorn Slough National Estuarine Research Reserve and Monterey Bay National Marine Sanctuary represent critical coastal ecosystem and habitat areas adjacent to the watershed's outlet.

Figure 1.2.2: Salinas River Watershed location along the larger Salinas River reach. The orange oval indicates the lower Salinas River Watershed focus of this management plan, which is only a subsection of the larger Salinas River. Source: Wikimedia.

1.3 Topography

The Salinas Valley is a broad valley formed by the Salinas River, running approximately 90 miles towards the NW with two bounding mountain ranges; the Santa Lucia Range on the west and the Gabilan Range on the east. **Figure 1.3.1** highlights how the Salinas River drains approximately 4,600 square miles of land in Monterey and San Louis Obispo Counties, emptying into the Monterey Bay (RCDMC). The Santa Lucia Range's highest summit, Junipero Serra Peak, stands 5,857 feet. In the upper watershed, the drainage rises to a maximum elevation of approximately 2,800 feet above mean sea level with steep topography characterizing much of the western portion (USGS, 2019).

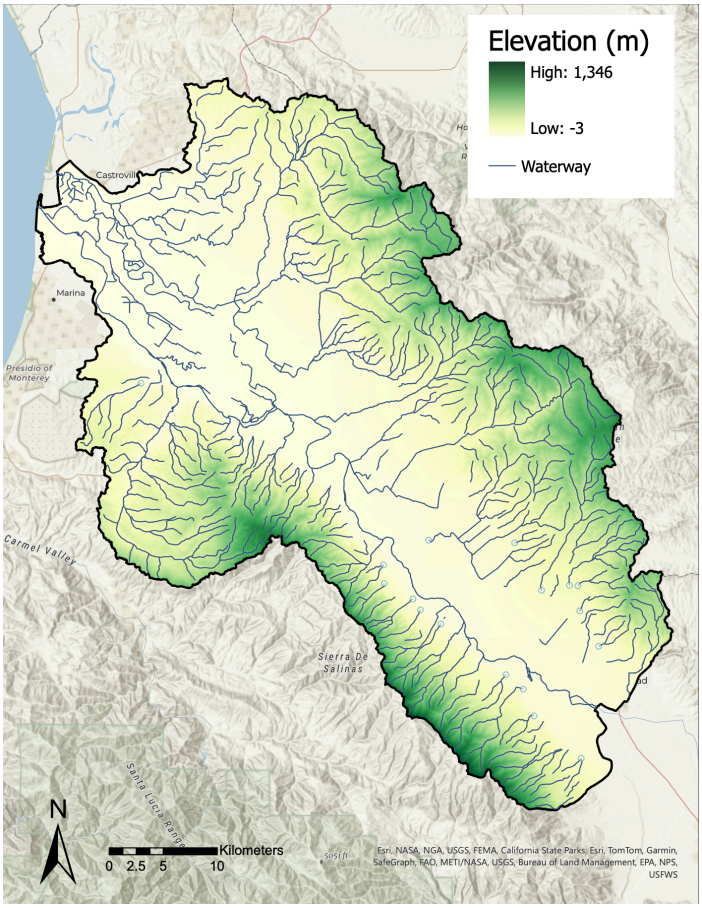

Figure 1.3.1: Topography of the Salinas River watershed. The watershed is bordered by the steep Santa Lucia and Gabilan Mountain Ranges, with a flat low-elevation valley floor.

1.4 Climate

The Salinas Watershed in California's central coast has a Mediterranean climate with mild, wet winters and warm, dry summers. Average annual temperatures range from 56.9°F to 57.5°F (13.8–14.2°C), with cooler coastal areas near Prunedale and North Salinas and warmer inland areas around Gonzales and Chualar. Seasonal temperature changes are moderate, with the highest temperatures usually in July and August and the lowest in December and January. Long-term temperature data from the PRISM Climate Group (1950–2024) show notable year-to-year variability and a gradual warming trend of about 1°F over the period (Figure 3.1). Annual rainfall averages 11–17 inches (279–432 mm), with more rain along the coast and less in inland valleys. Most rain falls between November and March, while summer tends to be dry (Figure 3.2).

These climate patterns which are shown in **Figure 1.4.1** and **Figure 1.4.2** significantly affect water processes in the watershed, including streamflow, groundwater recharge, and agricultural water needs. The clear wet–dry seasonal cycle and warming trend emphasize the need for flexible water management across the 4,650-square-mile basin (PRISM Climate Group, Oregon State University, 2024).

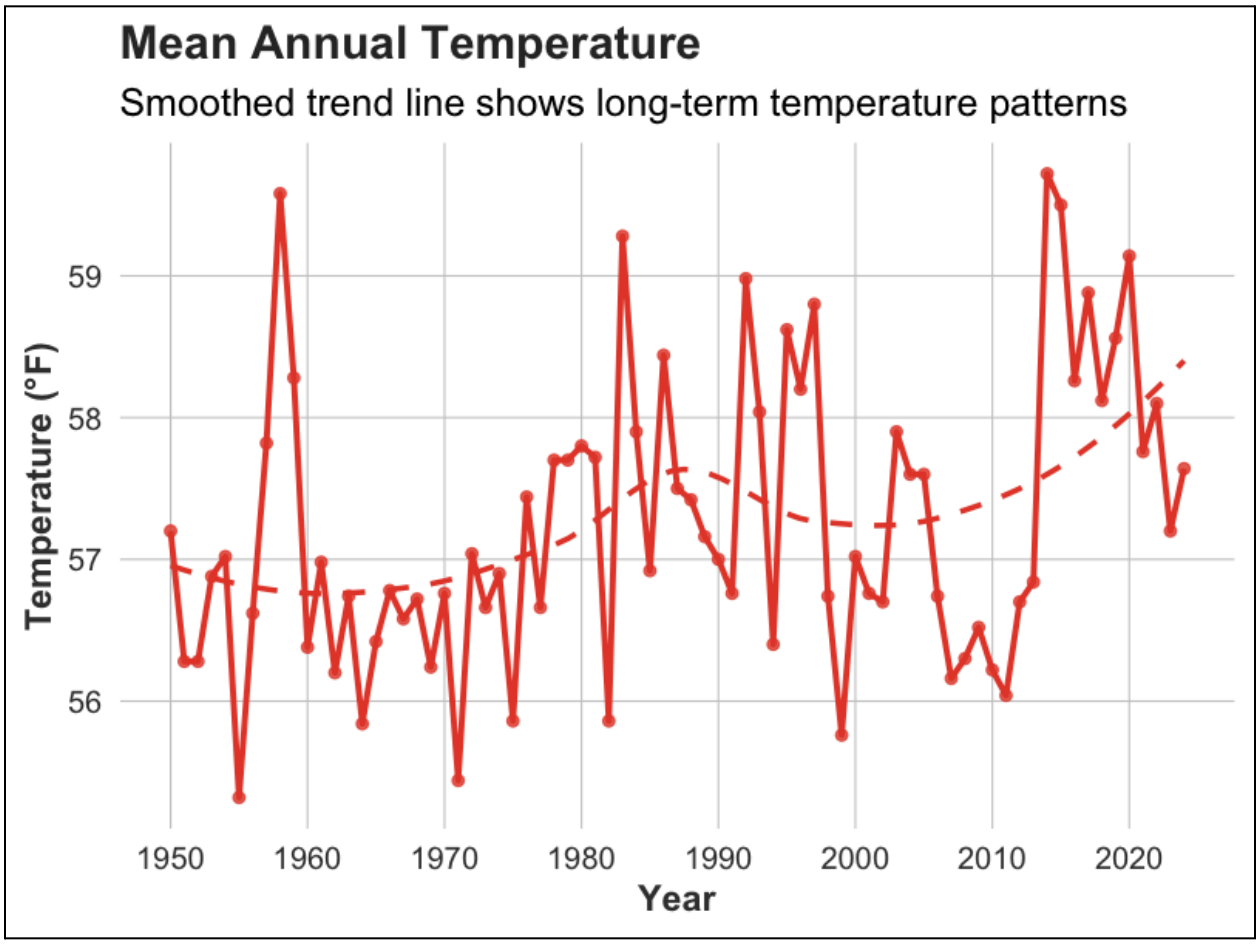

Figure 1.4.1: Mean Annual Temperature (1950–2024). Mean annual air temperature (°F) showing year-to-year variation and long-term warming patterns across the study period. The dashed line represents a smoothed trend highlighting gradual changes in temperature over time. Data were obtained from the PRISM Climate Group, Oregon State University (http://www.prism.oregonstate.edu/).

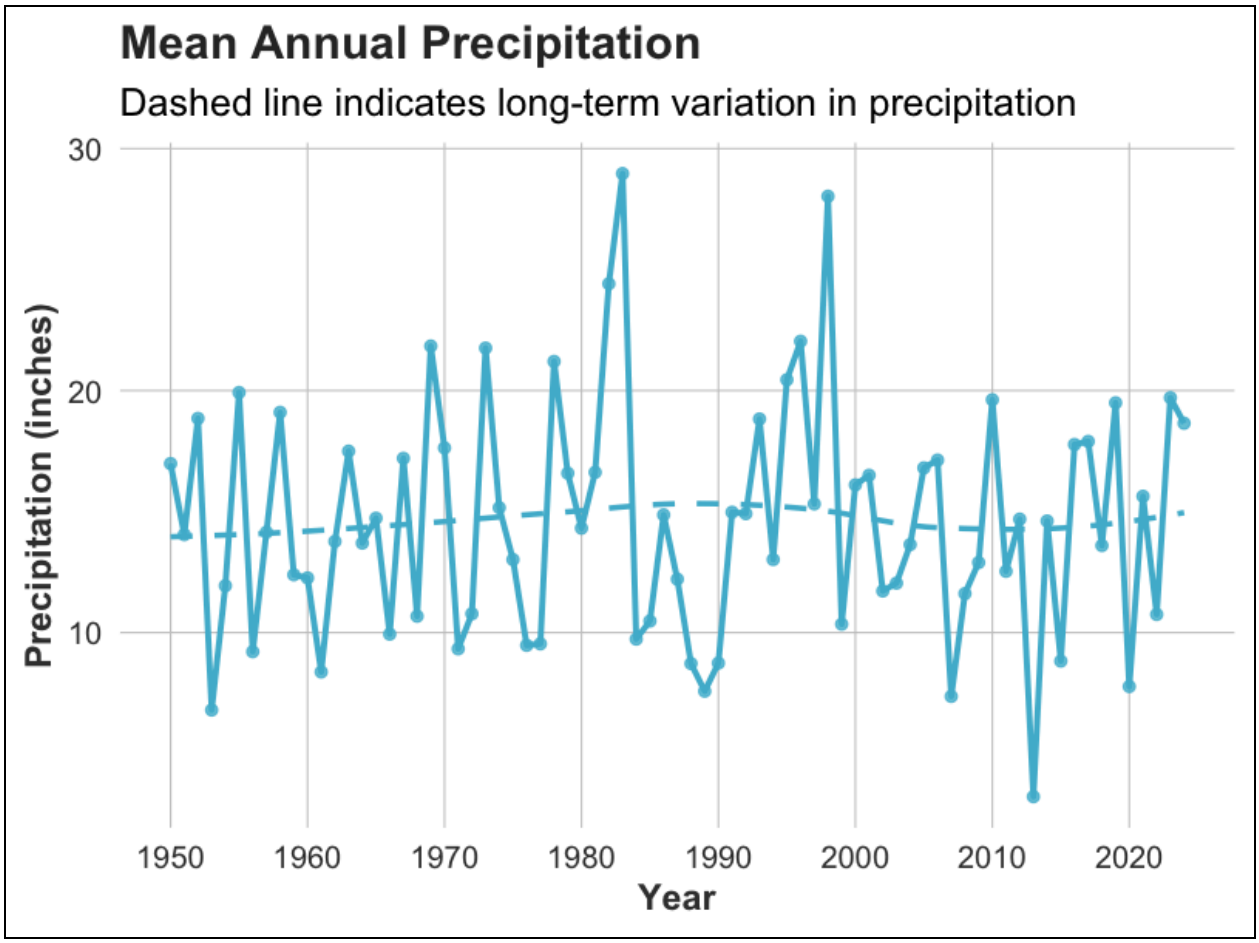

Figure 1.4.2: Mean Annual Precipitation (1950–2024). Mean annual precipitation (in inches) showing interannual variability across the study period. The dashed line represents the long-term trend in precipitation. Data were obtained from the PRISM Climate Group, Oregon State University (http://www.prism.oregonstate.edu/).

1.5 Geology

The Salinas River watershed falls within the Coast Range geomorphic province of California, which is composed of thick sedimentary strata. The Salinas Valley is a large intermontane valley that extends southeastward from Monterey Bay to Paso Robles, running subparallel to the San Andreas Fault, and the coastline is uplifted, terraced, and wave cut (CA Coastal Commission, 2002). The Salinas Valley is filled with thousands of feet of marine and terrestrial sediments overlying a granitic basement. The Salinas Valley Groundwater Basin is a layered aquifer system with aquitards creating distinct aquifers that range from 180 foot unconfined aquifers consisting of sand and gravel lenses up to deep 900 foot aquifers of alternating sand gravel mixtures and clays. While rare, these deeper aquifers are occasionally developed near the coast because seawater intrusion affects the shallower aquifers (CA Dept of Water Resources, 2004). Public supply wells are typically drilled to depths of 200 to 650 feet, consisting of solid casing from the land surface to depths of about 175 to 500 feet (USGS, 2011). Agricultural land in the Eastern Salinas Valley has moderate to excellent permeability, allowing rainfall to infiltrate soils and

recharge groundwater. **Figure 1.5.1** highlights the distribution of hydrologic soil groups across the watershed, showing how differences in infiltration potential shape groundwater recharge patterns.

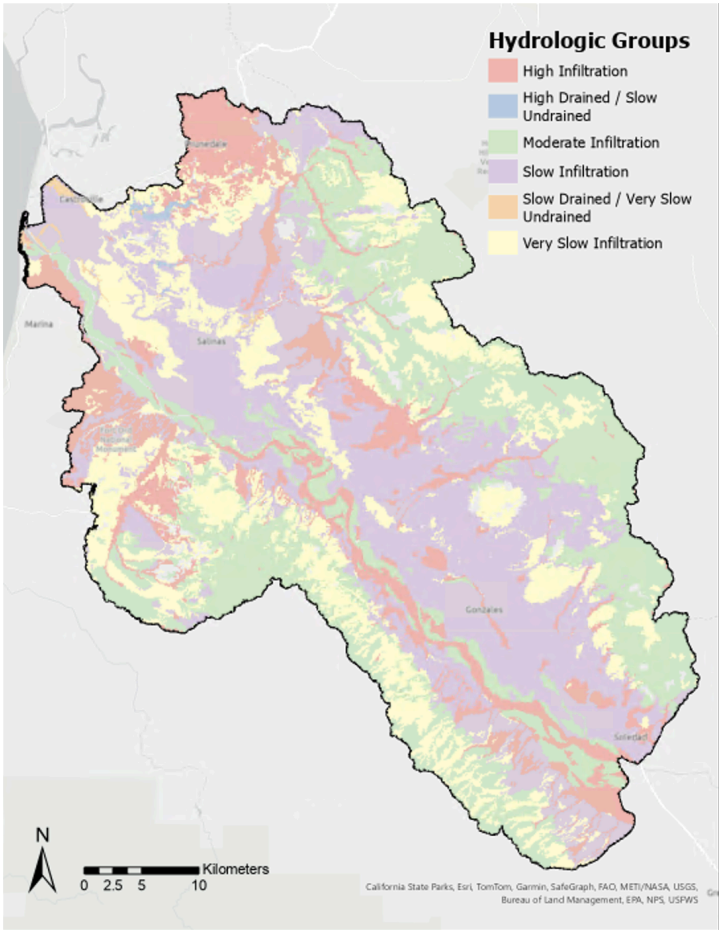

Figure 1.5.1: Distribution of hydrologic soil groups that make up the Salinas Watershed. From High Infiltration (red) to Very Slow Infiltration (blue). Hydrologic soil groups classify soils based on their infiltration and runoff potential. Group A soils (low runoff potential) dominate the mountainous uplands and foothills, while Group B and C soils are prevalent in the valley floor and intermediate elevations. Group D soils (high runoff potential) are scattered throughout the watershed. The distribution of soil groups influences surface water runoff, groundwater recharge rates, and flood risk across the watershed.

1.6 Soil Characteristics

Soil properties are fundamental controls on water infiltration and erosion processes within the Salinas Watershed. Soil permeability, shown in **Figure 1.6.2**, measures the rate at which water moves through soil and directly determines how much precipitation infiltrates into the ground versus becoming surface runoff. Highly permeable soils in the upland areas and foothills of the watershed facilitate rapid infiltration and groundwater recharge, allowing water to reach aquifer

systems that support the region's agricultural and municipal water demands. In contrast, lower-permeability soils on portions of the valley floor restrict infiltration and promote surface runoff, which can increase streamflow and lead to flooding during precipitation events. The spatial distribution of soil permeability across the watershed, therefore, shapes the balance between groundwater storage and surface water generation, influencing both water availability and flood risk.

Soil erodibility, illustrated in **Figure 1.6.1,** is the susceptibility of soils to water erosion and is determined by soil properties such as texture, structure, permeability, and organic matter content. Steeper terrain in the mountainous regions and Santa Lucia foothills generally exhibits higher erodibility due to slope and exposed soils, while valley floor soils typically show lower to moderate erodibility. Areas of high erodibility are vulnerable to accelerated erosion from stormwater runoff and concentrated flow, which mobilizes soil particles and increases sediment loads in streams and downstream water bodies. This sediment transport degrades water quality and can reduce the capacity of reservoirs and channels. Together, the patterns of soil permeability and erodibility in the Salinas Watershed create distinct hydrologic and erosional regimes across the landscape, with upland soils promoting infiltration and erosion, and valley soils affecting the timing and character of streamflow and sediment delivery.

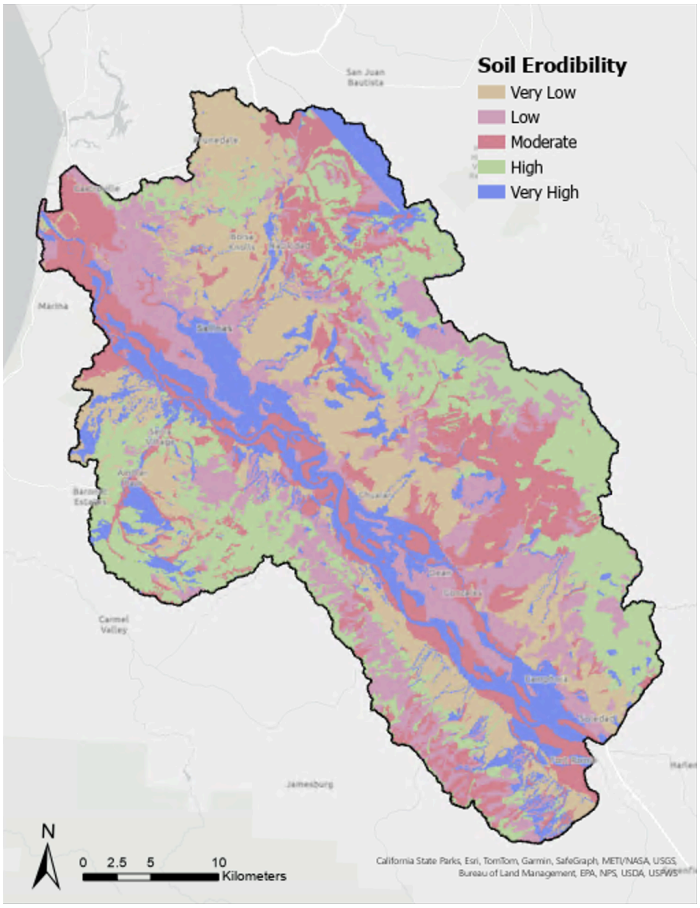

Figure 1.6.1: Distribution of soil erodibility for the Salton Sea Watershed. Very High > 0.45; High 0.35–0.45; Moderate 0.25–0.35; and Low-Very Low < 0.2. Soil erodibility indicates the susceptibility of soils to erosion by water based on soil properties such as texture, structure, permeability, and organic matter content. Steeper terrain in the mountainous regions and Santa Lucia foothills generally exhibits higher erodibility due to slope and exposed soils, while valley floor soils typically show lower to moderate erodibility. Areas of high erodibility are vulnerable to accelerated erosion from stormwater runoff and concentrated flow, which can impact water quality and sediment loads in streams and downstream water bodies.

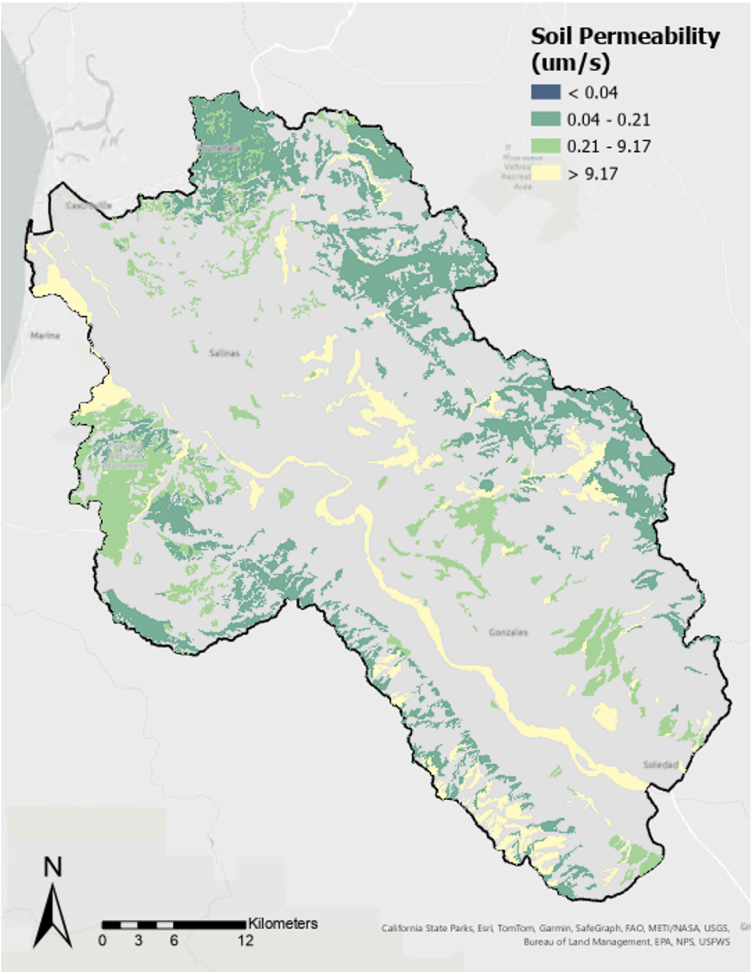

Figure 1.6.2: Distribution of Soil Permeability for the Salinas River Watershed. Very slow permeability is < 0.035 µm of water moving through the soil per second, Slow permeability is 0.035-0.210 µm of water per second, Moderate permeability is 0.21-9.17 µm of water per second, and Rapid permeability is over 9.17 µm of water per second. Soil permeability measures the rate at which water moves through soil and is a critical factor influencing groundwater recharge and surface runoff generation. Highly permeable soils in the upland areas and foothills facilitate infiltration and groundwater recharge, while lower permeability soils in portions of the valley floor tend to generate greater surface runoff and may restrict infiltration. The spatial variation in soil permeability across the watershed affects water movement through different soil layers and influences both the timing and volume of water reaching groundwater aquifers and surface water systems.

1.7 Land-Use

The Salinas Watershed encompasses a mosaic of land uses that reflect both intensive agricultural production and surrounding natural landscape features, as shown in **Figure 1.7.1.** The valley floor from Salinas through Gonzales to Soledad is dominated by cropland and pastureland, with approximately 200,000 acres of irrigated agriculture producing the majority of the nation's lettuce, broccoli, artichokes, strawberries, and cauliflower (RCDMC). This concentration of irrigated agriculture makes the Salinas Valley the "Salad Bowl of the Nation" and serves as the region's economic engine. Beyond the valley's intensive agricultural core, the

watershed transitions to rangelands and oak woodlands in the surrounding foothills. In contrast, coniferous forests and mixed woodlands characterize the higher elevations of the bordering Gabilan and Santa Lucia mountain ranges. Scattered urban centers, including Salinas, Gonzales, and Soledad, occupy relatively modest portions of the valley floor but serve as important regional hubs for commerce, services, and residential activity (Visión Salinas).

Historically, the Salinas Watershed was characterized by extensive rangelands, grasslands, and riparian wetlands before intensive agricultural development in the early 20th century transformed the valley floor into one of the nation's most productive agricultural regions (Monterey County Farm Bureau). This conversion dramatically reshaped the landscape, replacing native vegetation and natural water bodies with irrigated cropland and associated infrastructure. Today, urban development in Salinas, Gonzales, Soledad, and surrounding communities continues to expand incrementally as population growth and economic activity intensify. Future projections indicate that while agricultural production will likely remain the dominant land use in the valley, urban and suburban development is expected to gradually increase in and around existing towns, potentially converting some peripheral agricultural land to residential and commercial uses (Visión Salinas). Additionally, climate change and sustainability concerns may drive shifts toward less water-intensive crops in some areas. However, the region's established agricultural infrastructure and market advantages are likely to sustain the dominance of current crop production patterns in the foreseeable future.

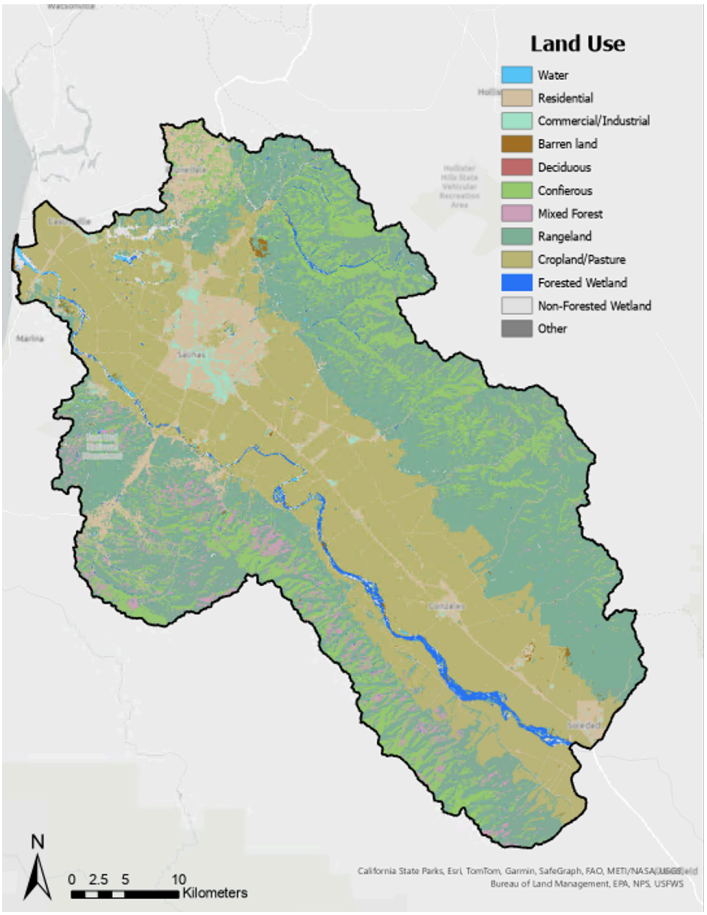

Figure 1.7.1: Multiple Land Use Designations for Salinas Watershed. The east and West borders of the watershed are made up of Coniferous forests and Rangelands, converging onto Cropland/Pasture. A Forested Wetland runs through much of the agricultural area into the coast. Residential and Commercial/Industrial space is concentrated around the city of Salinas, with some sprawl continuing into the surrounding mountains. Source: National Land Cover Database (NLCD) 2019 Products (ver. 3.0, February 2024)

1.8 Hydrology

The Salinas River is the longest river of California's Central Coast region, running 175 miles and draining 4,160 mi², with the watershed encompassing 4,530 mi² when including coastal drainages adjacent to the Salinas River outflow. The watershed's hydrology is characterized by a Mediterranean climate, with simulated evapotranspiration representing the largest component of the water budget after precipitation, averaging 13.9” per year compared to basin-wide average precipitation of 18.4” per year (USGS, 2019). Groundwater resources serve as the primary water source for crop irrigation and municipal supply, with two large multipurpose reservoirs also supplying irrigation and municipal water uses. Primary groundwater recharge comes from percolation in Arroyo Seco and Salinas River stream channels, with additional recharge from applied irrigation water. Seawater intrusion has advanced since the 1940s due to chronic groundwater overdraft conditions, with elevated salinity levels recorded less than half a mile from the city of Salinas by 2014 (Monterey County Water Resources Agency). The simulated

71-year average streamflow at the mouth of the Salinas River was only about 190 fps, indicating that most streamflow generated in the basin is lost to channel seepage. The watershed includes 200,000 acres of irrigated agriculture and features significant ecological challenges. **Figure 1.8.1** displays the various tributaries and waterways that connect the upland mountain ranges to the agricultural valley floor, eventually either infiltrating the groundwater basins or emptying into Monterey Bay.

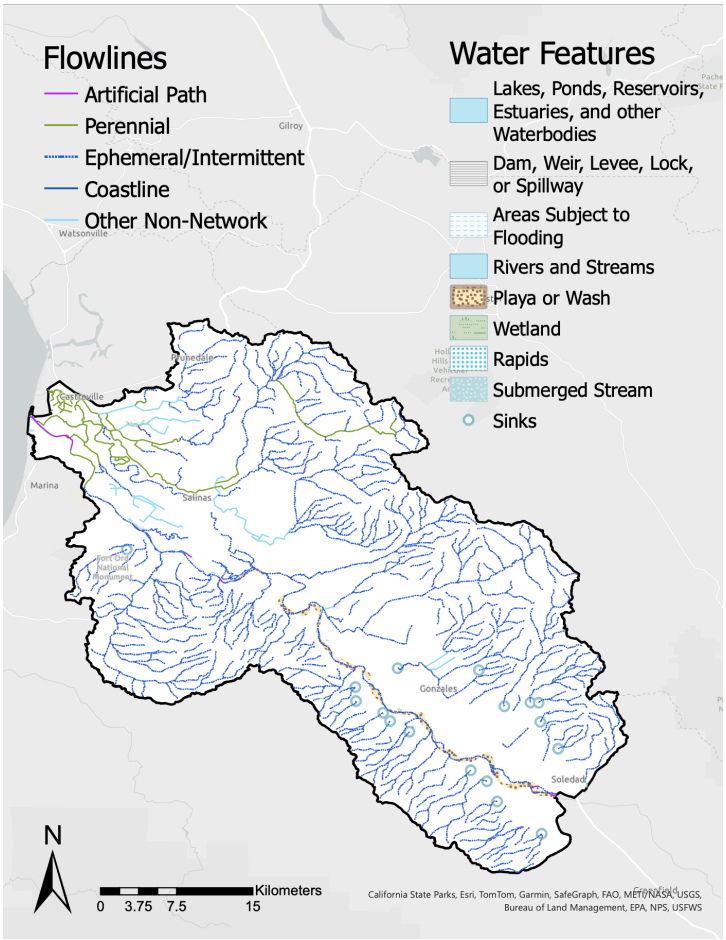

Figure 1.8.1: Major water features of the Salinas River watershed. Flowlines indicate the artificial and natural waterways, smaller upland streams feed into the larger Salinas River and accompanying tributaries in the valley floor. Source: BASINS 4.0

1.9 Biological Assets

The Salinas Valley and Monterey Bay region contain some of California's most ecologically diverse coastal ecosystems. The Elkhorn Slough, near Moss Landing, is one of the state's most extensive remaining wetlands and supports over 340 bird species, 100 fish species, and many invertebrates (Elkhorn Slough Foundation, 2025). Designated as a National Estuarine Research Reserve, it serves as an essential stopover along the Pacific Flyway for migratory shorebirds and waterfowl (National Audubon Society, 2025).

Surrounding areas include coastal dunes, salt marshes, and riparian corridors that provide critical habitat for several threatened and endangered species, including the Western snowy plover (Charadrius nivosus nivosus), California red-legged frog (Rana draytonii), tidewater goby (Eucyclogobius newberryi), and Monterey spineflower (Chorizanthe pungens var. pungens) (California Department of Fish and Wildlife [CDFW], 2025). The steelhead trout (Oncorhynchus mykiss irideus) relies on the Salinas River as a key migration and spawning corridor connecting inland streams to the ocean (California Natural Resources Agency [CNRA], 2023).

Along the coast, the Monterey Bay National Marine Sanctuary protects kelp forests, rocky reefs, and sandy-bottom habitats that support sea otters (Enhydra lutris nereis), harbor seals (Phoca vitulina), and commercially valuable fisheries (National Oceanic and Atmospheric Administration [NOAA], 2024). These resources sustain both biodiversity and local economies.

However, intensive agriculture and groundwater extraction contribute to nutrient pollution and habitat loss in the Slough and Salinas River. Climate change and sea level rise further threaten wetlands and dunes, altering salinity and reducing habitat for sensitive species (U.S. Fish and Wildlife Service [USFWS], 2025). Together, these coastal and riparian systems represent vital biological assets that maintain ecological balance and support community livelihoods.

Western Snowy Plover (Charadrius nivosus nivosus)

California Red-Legged Frog (Rana draytonii)

1.10 Summary of Watershed Characteristics

The Salinas River Watershed is located on California's Central Coast in Monterey and San Luis Obispo Counties, draining approximately 4,600 square miles before emptying into Monterey Bay. Positioned between the Santa Lucia and Gabilan mountain ranges, the watershed encompasses the productive agricultural valley floor as well as surrounding rangelands, forests, and critical coastal ecosystems. Understanding the watershed's geographic location within California's broader hydrologic and administrative landscape is essential for effective management planning and coordination with neighboring water systems and regulatory agencies.

Section 2: Population, Socioeconomic Conditions and Water Demand

2.1 Introduction

The Salinas River Watershed encompasses a diverse region shaped by agricultural production, rapid population growth, and complex water management challenges. Serving over 123,000 residents across five distinct public water systems in Monterey County, the Salinas District confronts the dual imperatives of meeting growing urban and agricultural water demands while maintaining long-term basin sustainability. The region's socioeconomic profile—characterized by a predominantly Hispanic and Latino population, median household income below the state average, and heavy reliance on agricultural labor—underscores the need for equitable water management strategies. With irrigation accounting for 44 percent of county-wide water withdrawals and an unsustainable dependence on groundwater to supply 70 percent of agricultural irrigation needs, the District faces mounting pressure to balance economic vitality with environmental resilience. This section examines the population dynamics, water use patterns, and supply characteristics that define the Salinas River Watershed, establishing the foundation for sustainable management practices and informed policy decisions through 2045.

Population and socioeconomic data has not been delineated by the exact boundaries of the watershed, so for this section trends and statistics for Monterey County as a whole and the Salinas District are considered. Monterey County provides a broader scope overview while Salinas District helps to inform urban trends specific to the watershed communities.

For the District level examination, the Salinas District 2020 Urban Water Management plan is utilized throughout this section. The District provides water service to portions of the City of Salinas and segments of unincorporated Monterey County lands. The service area encompasses five distinct Public Water Systems:

1. Salinas PWS
2. Country meadows Mutual PWS
3. Salinas Hills PWS
4. Las Lomas PWS
5. Oak Hills PWS

2.2 Population Trends

To first understand the larger regional context it is important to consider Monterey County as a whole, which encompasses the Salinas Watershed. Looking at the larger trend in Monterey County, there is a sharp population growth from 1990 to 2020. **Figure 2.2.1** shows a 22.6%

increase, with an average of 2,695 new residents per year and a total population of 432,858 in 2022.

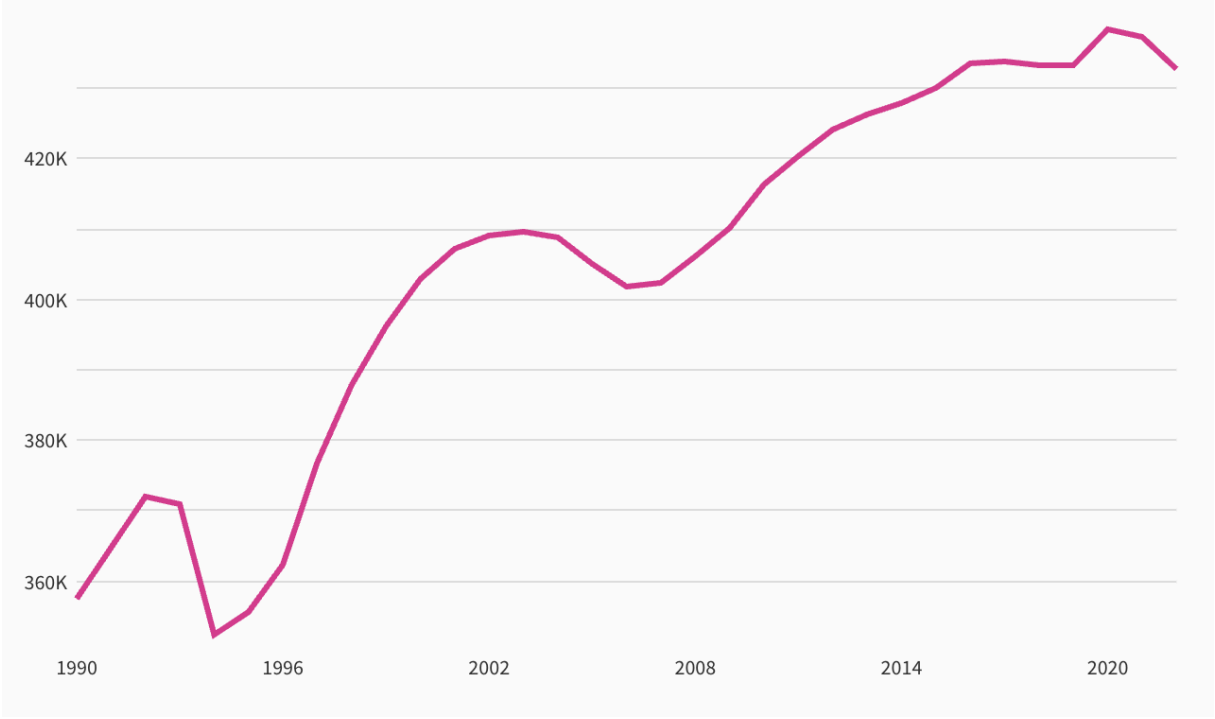

Figure 2.2.1: Monterey County Population Trends from 1990 - 2022. Monterey County's population has increased from 357,535 in 1990 to 432,858 by 2022. Source: USA Facts.

In the Salinas River Watershed, the three primary urban centers—Salinas, Gonzales, and Soledad—show relatively modest long-term growth trajectories. **Figure 2.2.2** presents census counts from 2010 and 2020 alongside population projections for 2030, 2040, and 2050 from AMBAG's Final Draft 2026 Regional Growth Forecast. Overall, Salinas is projected to increase by less than 10% between 2010 and 2050, Soledad remains generally stable with a slight long-term decline, and Gonzales exhibits the strongest relative growth at just over 40%. Despite this variation, these cities together still comprise less than half of Monterey County's total population, and the watershed is not expected to experience substantial urban expansion. As a result, urban growth is not a major driver of future water management challenges in this region. **Table 2.2.1** breaks down the population growth from present estimates (2023) and future projections (2050).

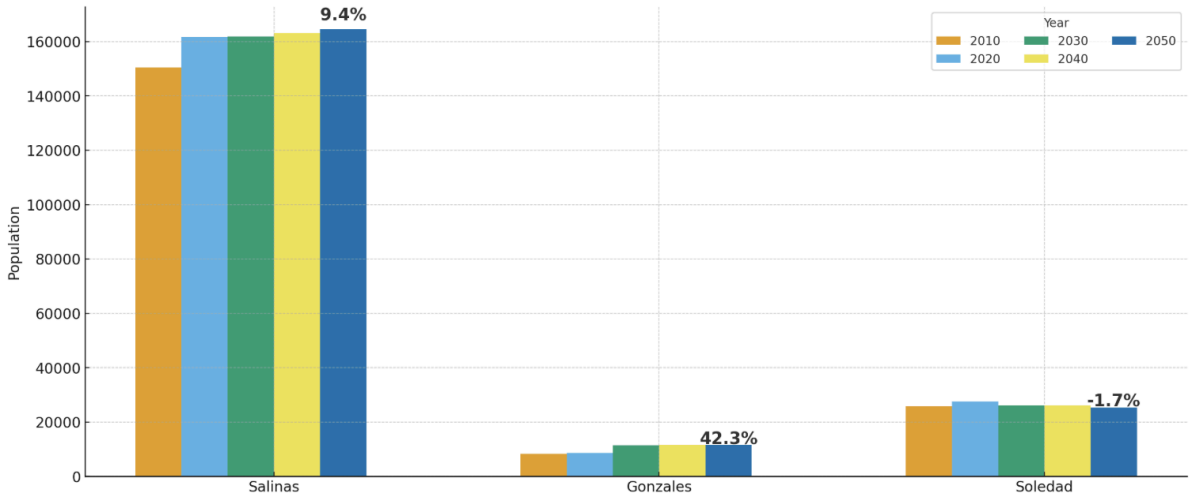

Figure 2.2.2: Population Trends in the Major Urban Areas of the Salinas River Watershed. Population trends for Salinas, Gonzales, and Soledad from 2010 to 2050. Bars show census counts and regional projections, with labels indicating cumulative percent change relative to 2010. Gonzales experiences the highest relative growth, while Salinas remains stable and Soledad shows a slight long-term decline.. Source: Association of Monterey Bay Area Governments (AMBAG).

Table 2.2.1: Current Population Estimates and Future Projections for Salinas, Gonzales and Soledad.

City	2023 Population	2050 Projected Population	Change
Salinas	160,870	164,615	+2%
Gonzales	8,384	11,647	+39%
Soledad	26,426	25,292	-4%

2.3 Socioeconomic Conditions

The socioeconomic profile of the Salinas District reflects both the region's agricultural roots and the persistent structural inequities that shape many working-class communities in California's Central Coast. As shown in Table 1, median household income in the Salinas District ($61,527) lags behind the statewide average ($75,235), while the share of residents living below the federal poverty line (16.8%) exceeds California's rate (13.4%). These figures highlight the area's dependence on agricultural labor and low-wage employment, where seasonal work and limited access to higher-paying industries contribute to economic vulnerability. The comparatively lower income levels also intersect with linguistic and educational barriers identified in **Table 2.3.1**, where only 11.6 percent of adults hold a bachelor's degree or higher, and nearly one in five households is classified as limited English-speaking. Together, these metrics suggest that despite its vital role in California's agricultural economy, the Salinas District faces persistent challenges in upward mobility and workforce diversification.

Table 2.3.1: Urban Population Statistics from Salinas District Compared to California (2020).

Location	Median Household Income	Population below Federal Poverty Level
California	$75,235	13.4%
Salinas District	$61,527	16.8%

Demographically, the community's social fabric is deeply tied to its Hispanic and Latino population, which makes up nearly 80 percent of residents (**Table 2.3.2**) and is represented across multiple racial categories in **Figure 2.3.3**. The racial distribution—dominated by individuals identifying as *Some Other Race* (53.0 percent) and *White* (39.9 percent)—underscores how the federal census often inadequately captures Latino identity within conventional racial categories. This cultural and linguistic composition shapes the district's educational and economic outcomes, as a large share of residents navigate bilingual or multilingual households. Taken together, the demographic and socioeconomic data point to a community that is young, ethnically vibrant, and central to California's agricultural production, yet continues to confront barriers related to income inequality, education, and access to opportunity.

Table 2.3.2: Summary Demographic for the City of Salinas (ACS 5-year 2019).

Category	Value
Median Age (years)	38.5
Hispanic or Latino	79.8%
Bachelor's Degree or Higher	11.6%
English only or 'very well' at home	62.2%
Limited English-speaking households	19.5%

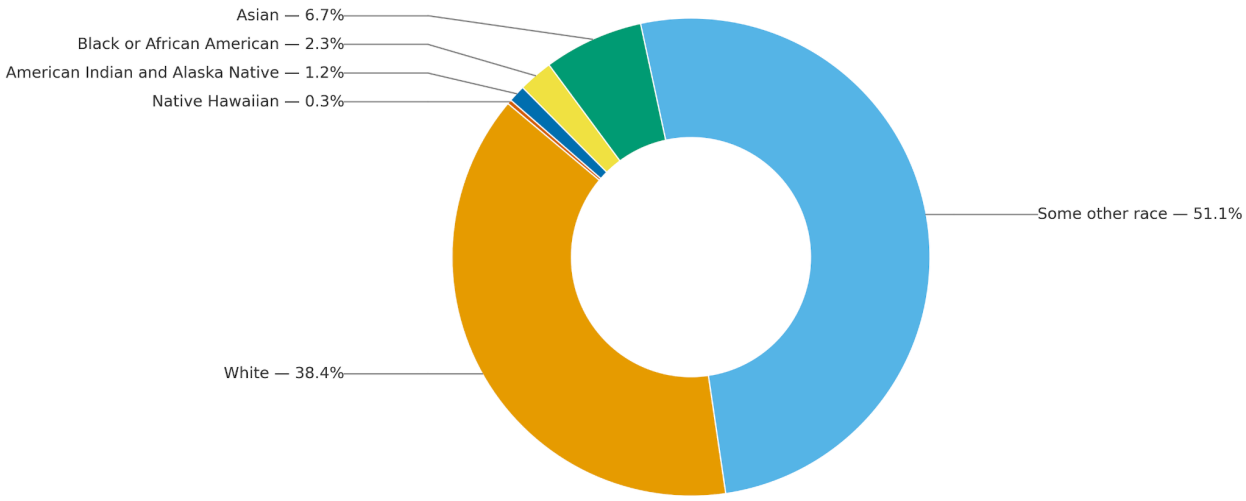

Figure 2.3.3: Racial Makeup of the City of Salinas (ACS 2019 5-Year Estimates). Racial composition of the City of Salinas based on data from the U.S. Census Bureau's 2019 American Community Survey (5-Year Estimates). The population is predominantly composed of individuals identifying as *Some Other Race* (53.0%) and *White* (39.9%), with smaller proportions identifying as *Asian* (7.0%), *Black or African American* (2.4%), *American Indian and Alaska Native* (1.2%), and *Native Hawaiian* (0.3%). These demographics reflect Salinas's strong Hispanic and Latino heritage, which spans multiple racial categories and accounts for the majority of the city's residents. Source: U.S. Census.

2.3 Water Demand

Water demand in the Salinas River Watershed varies among sectors, reflecting the region's strong agricultural foundation and limited surface-water resources. As shown in **Figure 2.3.1**, crop irrigation remains the primary water-use category in Monterey County, making up 44% of total withdrawals in 2015, followed by thermoelectric power generation and public supply. **Table 2.3.1** offers a detailed breakdown of total withdrawals, which amounted to approximately 425,000 acre/ft/yr in 2015. Mining, domestic, industrial, and livestock uses together accounted for less than 10% of total demand (California Department of Water Resources, 2024).

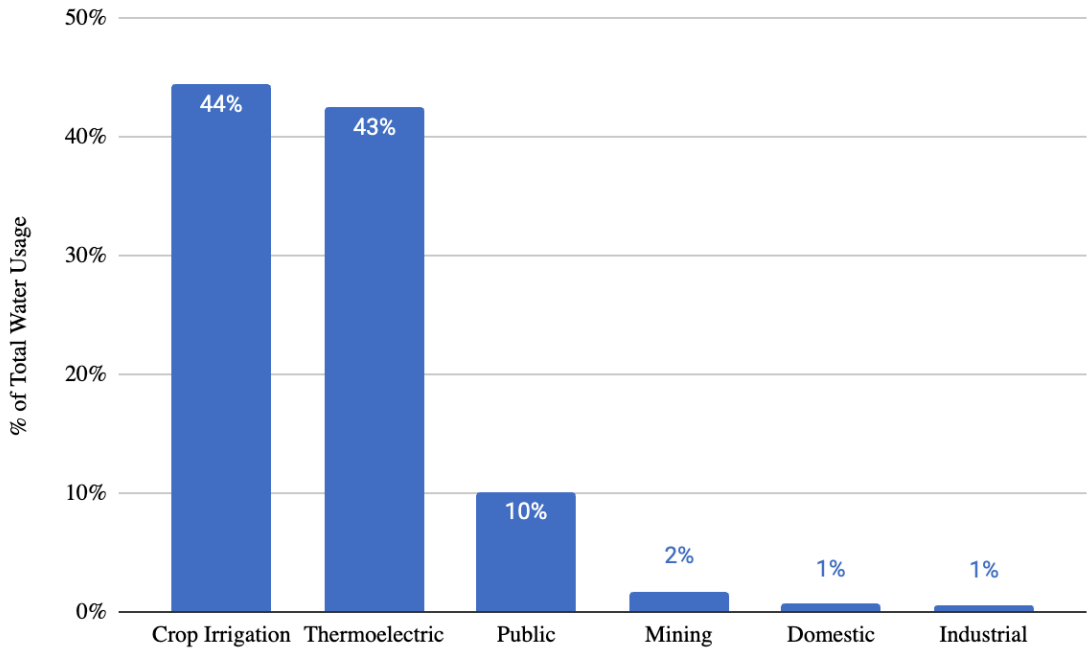

Figure 2.3.1: Monterey County Water Use By Category (2015). Proportion of withdrawn water for the following uses: crop irrigation (44%), thermoelectric power (43%), public (10%), mining (2%), domestic (1%), and industrial (1%). Aquaculture and golf irrigation water withdrawals are excluded because no water is used for that purpose. Figure based on data in Table 2. Source: U.S. Geological Survey data release (2018).

Within the urban Salinas District, water use has steadily declined since the early 2000s, as depicted in **Figure 2.3.2**, which shows a long-term decrease in per-capita demand and relatively stable projections through 2045 under normal-year and dry-year conditions. This decline reflects the combined effects of statewide conservation initiatives, technological efficiency improvements, and the adoption of drought-resilient landscaping practices (California Department of Water Resources, 2024).

Although municipal efficiency has improved, agricultural irrigation remains the primary driver of water use. With about 200,000 acres of irrigated cropland and 188,017.52 acre-ft/yr in water use for crop irrigation, shown in **Table 2.3.1**, growers are applying about 0.94 ft/acre/ yr of irrigation. This is in line with typical crops of vegetables, berries and vineyards, according to the Association of Monterey Bay Governments (AMBG) which typically require 0.8 to 1.5 feet of applied water annually depending on irrigation method and season length.

As compared to statewide patterns, this level of irrigation is relatively efficient. Across California, average applied irrigation depths typically range from 1.5 to 3.5 ft/acre/yr, with higher values associated with water-intensive crops such as nuts, alfalfa and summer field crops. The Salinas Valley's cooler coastal climate, shorter growing seasons and widespread adoption of drip irrigation contribute to significantly lower applied water requirements than many interior

basins. As a result, the estimated 0.94 ft/acre/yr is both reasonable for the local crop mix and notably below the statewide average for agricultural irrigation.

While these numbers show promise, the Salinas Valley Basin Groundwater Sustainability Plan identifies irrigation efficiency, groundwater recharge, and recycled-water expansion as essential measures to reduce overall demand and achieve basin-wide sustainability. Ongoing data integration efforts through the California Open Data Portal (2024) further support transparent water-use tracking across agricultural and urban sectors, enabling adaptive management and more informed decision-making for the region's long-term resilience.

Table 2.3.1: Monterey County Water Use by Category (2015). Units: acre-feet per year. Source: U.S. Geological Survey data release (2018).

Water Use Category	Total Withdrawals (acre-ft / yr)
Crop Irrigation	188,017.52
Domestic	2,734.96
Golf Irrigation	1,255.39
Industrial	2,421.12
Livestock	470.77
Mining	7,409.06
Public	4,2851.49
Thermoelectric	180,081.64
Total	**425,241.95**

Urban Water Demand

The urban water demand in the Salinas District, illustrated in **Figure 2.3.2**, shows the successful reduction of water consumption from peak usage in the early 2000s. Despite continuous population growth from 2004 to 2020, actual water use decreased by 21% from the historical peak. Per capita urban water consumption within the Salinas District has also become more efficient, with its current daily water usage at 44.8 residential gallons per day according to an analysis of the State Water Control Board's 2017 data. When compared against recent statewide average per capita urban water consumption rates of 77 gallons per day, this reflects a 40% lower daily consumption level driven by the pull of higher public awareness of water scarcity and push of conservation measures within the district (Greene, LA Times). Effective tiered consumption pricing, enhanced conservation programs, and adoption of modern plumbing efficiency standards primarily drove the reduction.

Future demand projections, shown in **Figure 2.3.2**, indicate a persistent upward trend through 2045 across the three hydrologic conditions. Under normal hydrologic conditions (represented

by the green dashed line in **Figure 2.3.2**), demand is projected to increase to 18,853 AFY by 2045. Importantly, the District projects that total water use will increase by only 15 percent between 2020 and 2045, even though the population is anticipated to grow by 24 percent, demonstrating the moderating effect of efficiency measures. Under the more severe scenarios depicted in the figure, demand is forecast to rise: up to 19,464 AFY in a single dry year (orange dashed line) and to 19,842 AFY during an extended multi-year drought (red dashed line) by 2045. Given that populations are only projected to rise modestly in Salinas, drop in Soledad and rise relatively modestly in Gonzales, this signals a sharp increase in per capita consumption.

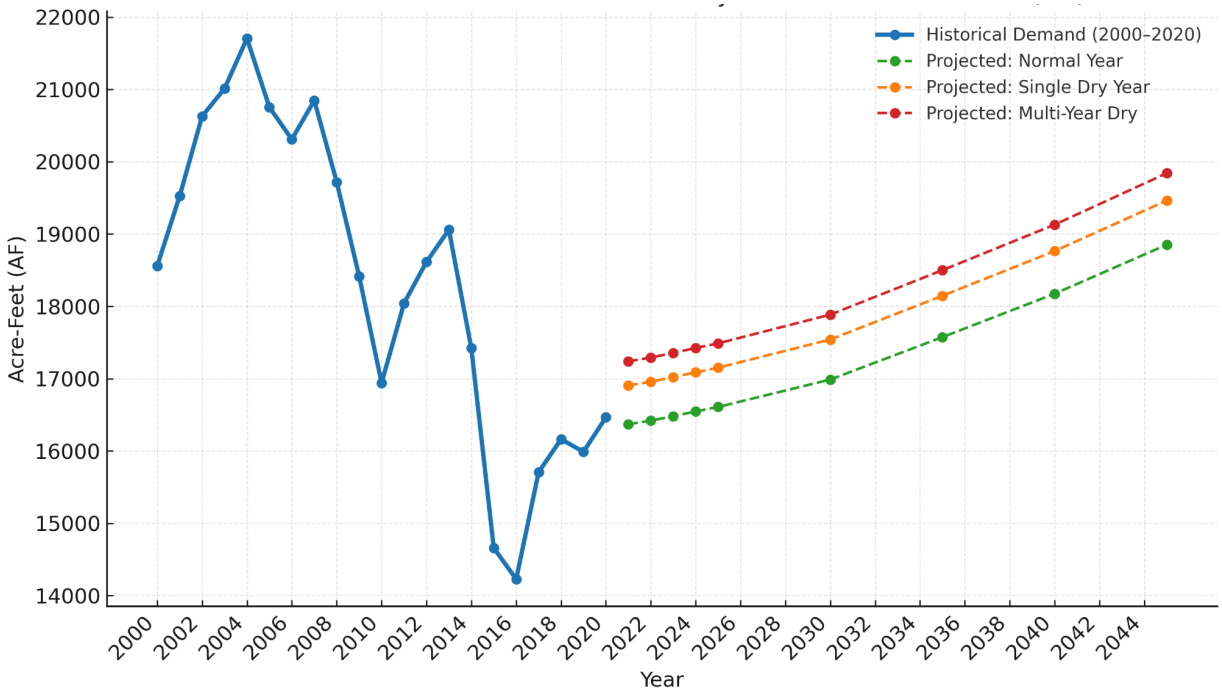

Figure 2.3.2: Historical and Projected Urban Water Demand for Salinas District (2000 - 2045). This chart illustrates total annual water demand (in acre-feet) over the past two decades and projections through 2045 under three hydrologic scenarios. Historical demand (blue line) shows a gradual decline from peak levels in the mid-2000s, reflecting conservation measures and efficiency improvements. Projections indicate relatively stable demand, with modest increases under all scenarios. The green dashed line represents normal-year conditions, the orange dashed line represents a single dry year, and the red dashed line reflects multi-year drought conditions with the highest expected demand. Source: City of Salinas 2020 Urban Water Management Plan.

2.4 Water Supply

Since 1962, California Water Services Company (Cal Water) has serviced the Salinas District PWS including Country Meadows, Salinas, Salinas Hills, Las Lomas, and Oak Hills from local water sources, as shown in **Figure 2.4.3b**. The majority of water within the Salinas River

Watershed is provided by surface water withdrawals (55% of total), with the rest coming from groundwater, as shown in **Figure 2.4.1**. However, irrigation water, which is the primary driver of withdrawals throughout the watershed, sources 70% from groundwater withdrawals – signaling a potentially unsustainable reliance on near-ocean groundwater stores. Interestingly, all 160.29 million gallons per day (Mgal/day) of water used for thermoelectric power is supplied by saline surface-water withdrawals. The same is true for mining surface-water withdrawal. As alluded to in this data from USGS, the Salinas River Watershed is situated near four seawater desalination facilities, from Moss Landing Power Plant on the northern end of Monterey Bay and the Sand City Desalination Plant at the south. While neither is directly located on the Salinas River, they have many potential implications for the nearby watershed. Desalination plants can help reduce demand for freshwater withdrawals; however, they also typically discharge highly concentrated brine byproducts back into the ocean, which could negatively impact marine ecosystems.

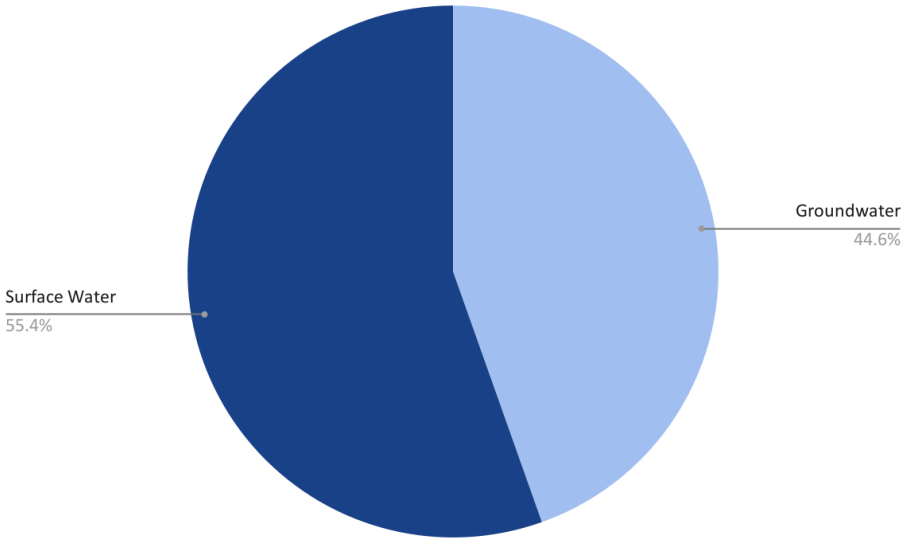

Figure 2.4.1: Total Groundwater vs. Total Surface Water Withdrawals for Monterey County (2015). Surface water accounts for 55% of total water withdrawal whereas groundwater represents 45%. Source: U.S. Geological Survey data release (2018).

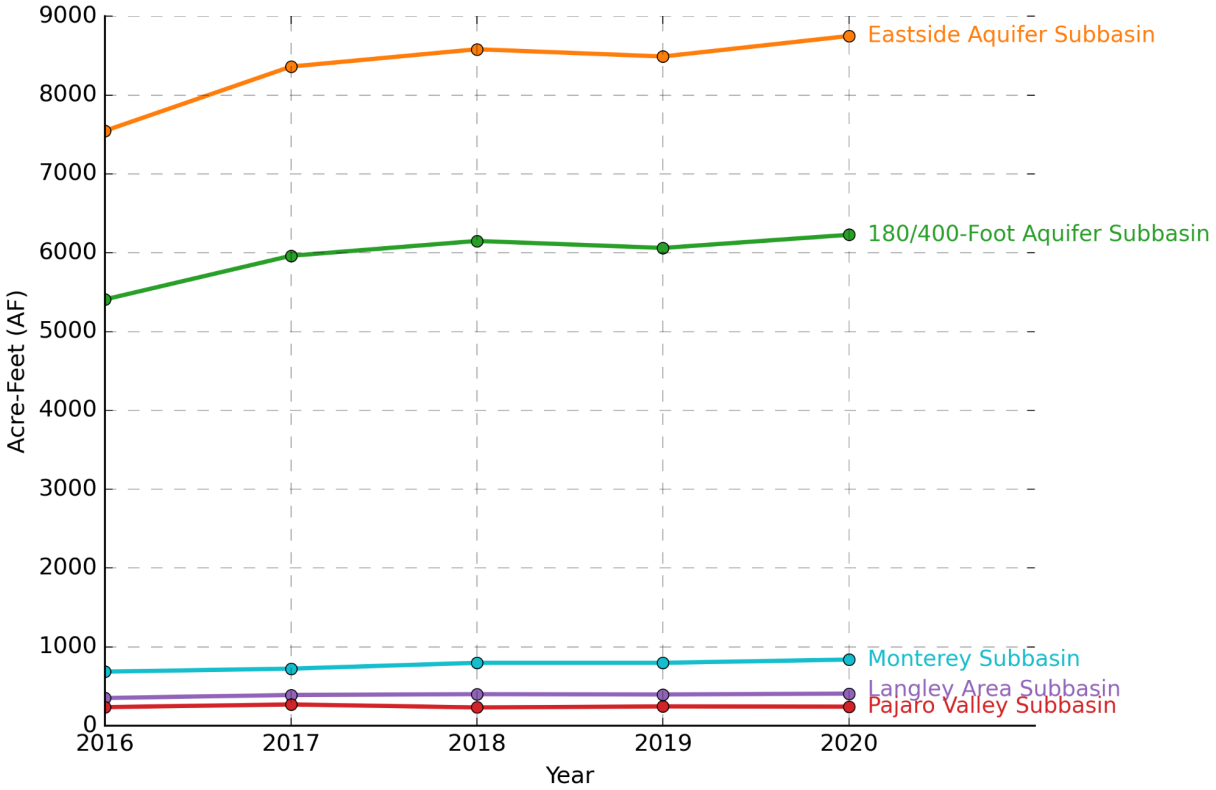

Figure 2.4.2a: Salinas District Groundwater Supply by Subbasins (2016-2020). Historical groundwater production across five major basins in the Salinas District between 2016 and 2020. Supply levels remained relatively stable during this period, with the Eastside and 180/400-Foot Aquifer Subbasins contributing the largest shares of total groundwater use. Minor year-to-year fluctuations reflect local hydrologic conditions and variability in agricultural and municipal pumping demands.

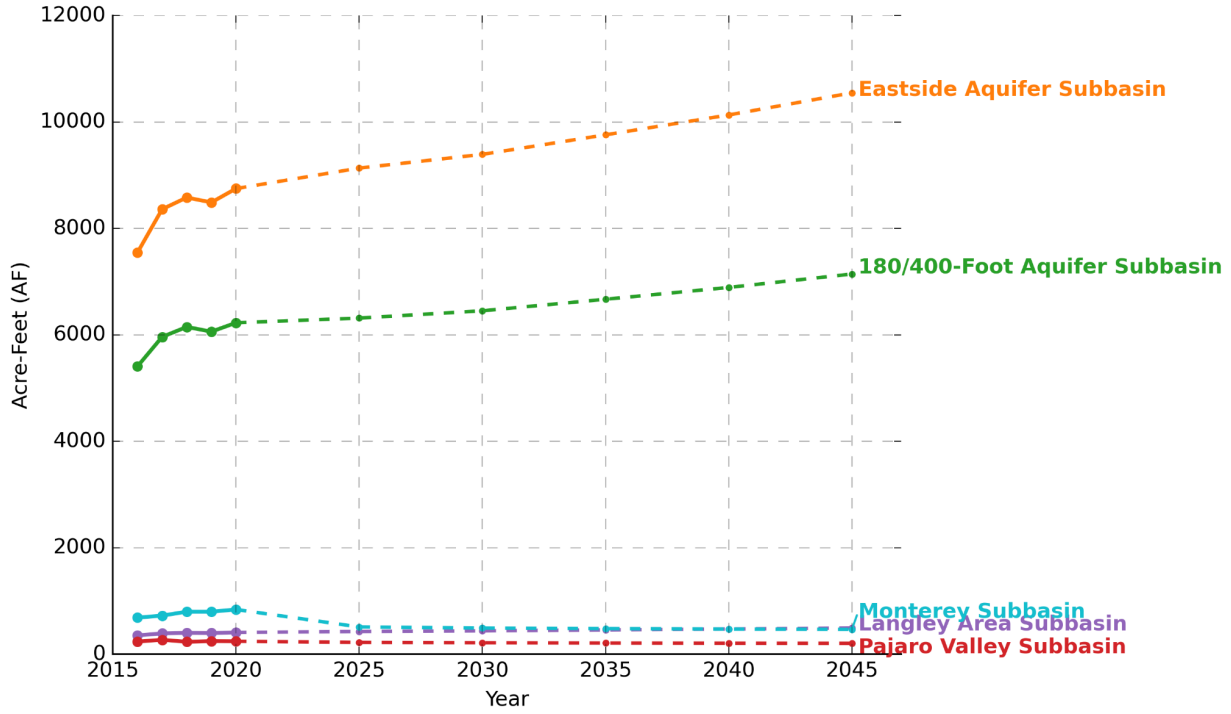

Figure 2.4.2b: Salinas District Water Supply Projections by Subbasins (2020-2045). Extending the historical groundwater supply trends to include projections through 2045. Solid lines represent measured supplies from 2016–2020, while dashed lines show projected availability under anticipated pumping and recharge conditions. Most basins are expected to experience gradual increases in supply, led by the Eastside and 180/400-Foot Aquifer Subbasins. The Monterey and Langley Area Subbasins remain relatively small contributors, while the Pajaro Valley Subbasin shows minimal change over time.

The Salinas Valley's municipalities, including Gonzales and Salinas (served by Cal Water), are entirely dependent on groundwater for municipal and industrial water supply, making groundwater sustainability a paramount concern for land use planning and watershed management. Gonzales and Cal Water both draw from the critically overdrafted 180/400-Foot Aquifer Subbasin. In response, groundwater management is governed by Groundwater Sustainability Plans (GSPs) under the Sustainable Groundwater Management Act (SGMA), with coordination between Cal Water and the Salinas Valley Basin Groundwater Sustainability Agency (SVBGSA).

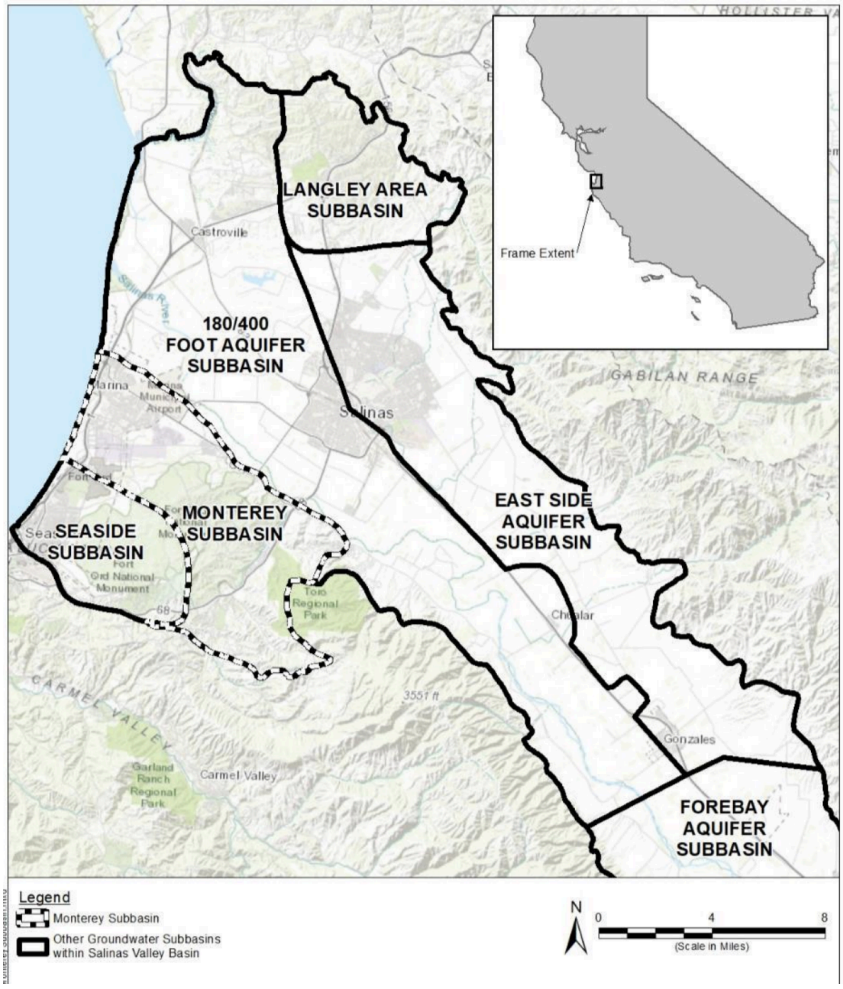

Figure 2.4.3a: Groundwater Subbasin designations throughout the Salinas River Watershed and beyond. The urban areas of the defined watershed are solely reliant on groundwater for urban drinking water supply. These basins have been heavily overdrafted for years, but now face stricter standards with the implementation of SGMA. Source: Monterey Subbasin Groundwater Sustainability Plan (2021).

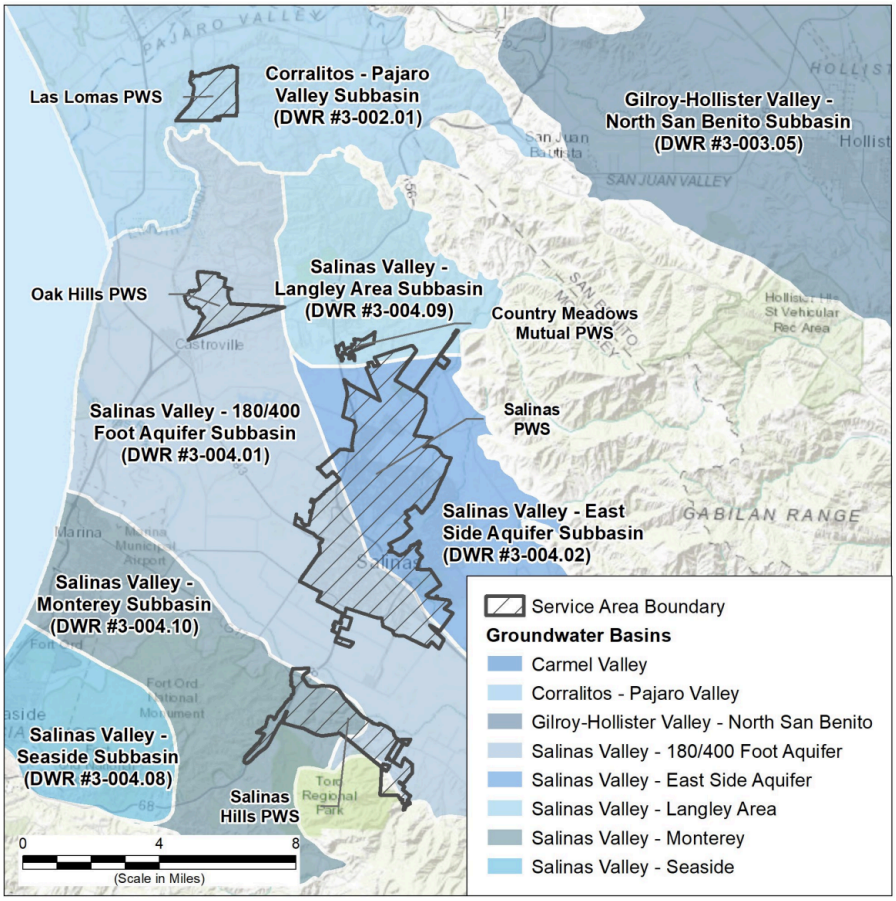

Figure 2.4.3b: Salinas Valley's Five Public Water Systems (PWS) superimposed over the Groundwater Basins. Source: City of Salinas 2020 Urban Water Management Plan.

Gonzales' General Plan establishes a critical policy objective: to maintain overall groundwater extraction at approximately 2010 usage levels (approximately 4.8 Mgal/day) upon buildout of the Urban Growth Boundary. This extraction cap ensures that new urban development—projected to add over 9,000 residential units across the three cities by 2045–2050—must be supported through water conservation, recycling, and reuse rather than increased groundwater pumping. The City encourages the use of gray water and rainwater cisterns for landscape irrigation in multi-family and commercial developments to reduce potable water demand. Gonzales and Cal Water both anticipate wastewater recycling as a key supply augmentation strategy, with the goal of deploying tertiary recycled water for irrigation and industrial uses.

Monterey County policy requires that new development requiring discretionary permits demonstrate proof of a long-term, sustainable water supply in both quality and quantity before approval is granted. This requirement directly constrains land use planning, as projected buildout cannot proceed without securing alternative water supplies or reducing demand through conservation and recycling. Water quality protection is also critical: elevated nitrate levels in parts of the groundwater basin (from agricultural operations) necessitate careful well

design, water treatment, and protection of natural drainages from hazardous materials through Best Management Practices (BMPs) to trap urban runoff pollutants.

2.5 Summary

The Salinas River Watershed serves over 123,000 residents across five public water systems in Monterey County, with population growth concentrated in the broader county rather than within watershed cities. The region's socioeconomic profile is characterized by a predominantly Hispanic and Latino population (80%), median household income below the state average ($61,527 vs. $75,235), and significant educational and linguistic barriers, reflecting dependence on agricultural labor and low-wage employment. Water demand is dominated by agricultural irrigation, which accounts for 44% of county-wide withdrawals and relies unsustainably on groundwater (70% of irrigation supply). Urban water use in the Salinas District has declined 21% since the early 2000s through conservation efforts, though demand is projected to increase modestly by 2045 due to new development—with total increases capped at 15% despite anticipated 24% population growth. The municipality and agricultural sectors are entirely dependent on groundwater from critically overdrafted basins, making groundwater sustainability and alternative water supplies (recycling, conservation) essential for supporting future development while maintaining basin viability.

Section 3: Land Use Planning

3.1 Introduction

Land use planning in the Central Salinas Valley represents a critical interface between urban growth management, agricultural preservation, and water resource sustainability. The region faces significant urban development pressures, driven partly by the relative affordability of housing compared to surrounding coastal areas, which creates tension with the imperative to conserve some of California's most productive agricultural lands. The Monterey County General Plan establishes a "City-Centered Growth" vision that channels new development to incorporated cities—Salinas, Soledad, and Gonzales—through compact, contiguous growth patterns while maintaining the distinction between urban and rural lands. This approach has direct implications for water demand, infrastructure requirements, and watershed management. Planning documents across these municipalities extend to 2045–2050, with specific plans and zoning frameworks providing the regulatory mechanisms to guide development while coordinating with water supply and demand projections through Urban Water Management Plans (UWMPs).

3.2 City of Salinas

The City of Salinas guides development through a General Plan paired with a comprehensive zoning framework organized by Place Types and anchored by two major Specific Plans that will substantially influence future water demand within the watershed.

The West Area Specific Plan (WASP), approved in 2019, encompasses 797 acres and is designed to accommodate more than 4,300 housing units, as well as mixed-use commercial areas, parks, and schools. The Central Area Specific Plan (CASP), approved in 2020, encompasses 760 acres and outlines plans for approximately 3,900 housing units, organized according to New Urbanism principles, featuring a mixed-use village center, schools, and parks. Together, these two specific plans represent over 1,500 acres of planned growth within the city. Their respective specific plan adoptions definitively establish land use and zoning for both areas and will not be subject to further modification unless significant plan amendments are made.

Salinas' zoning framework is organized into residential, commercial, and industrial districts with varying density designations, as depicted in the City of Salinas Land Use Map as shown in **Figure 3.2.1**. Residential zoning ranges from Low Density (R-L-5.5) at 12–20 dwelling units per acre to High Density (R-H) zones at 20–40 DU/Acre. Mixed-use areas, particularly along Main Street in the City Center place type, allow intensities from 1–8 stories with floor area ratios up to 6.0 FAR and residential densities reaching 30–80 DU/Acre. Industrial designations include Business Park, General Commercial, and General Industrial zones. This zoning structure supports infill and vertical development patterns, particularly in downtown areas, which may reduce per-capita water demand through higher density development.

Salinas' planning horizon extends to 2045, coordinated through the City's Urban Water Management Plan (UWMP), which is updated every five years and utilizes growth forecasts from AMBAG/Caltrans to project both water demands and supplies. This explicit linkage between land use assumptions and water planning ensures that the two WASP and CASP developments—totaling approximately 8,200 housing units—are accounted for in long-term water availability assessments. However, the materialization of these projections depends on timely infrastructure development and adequate surface and groundwater supplies.

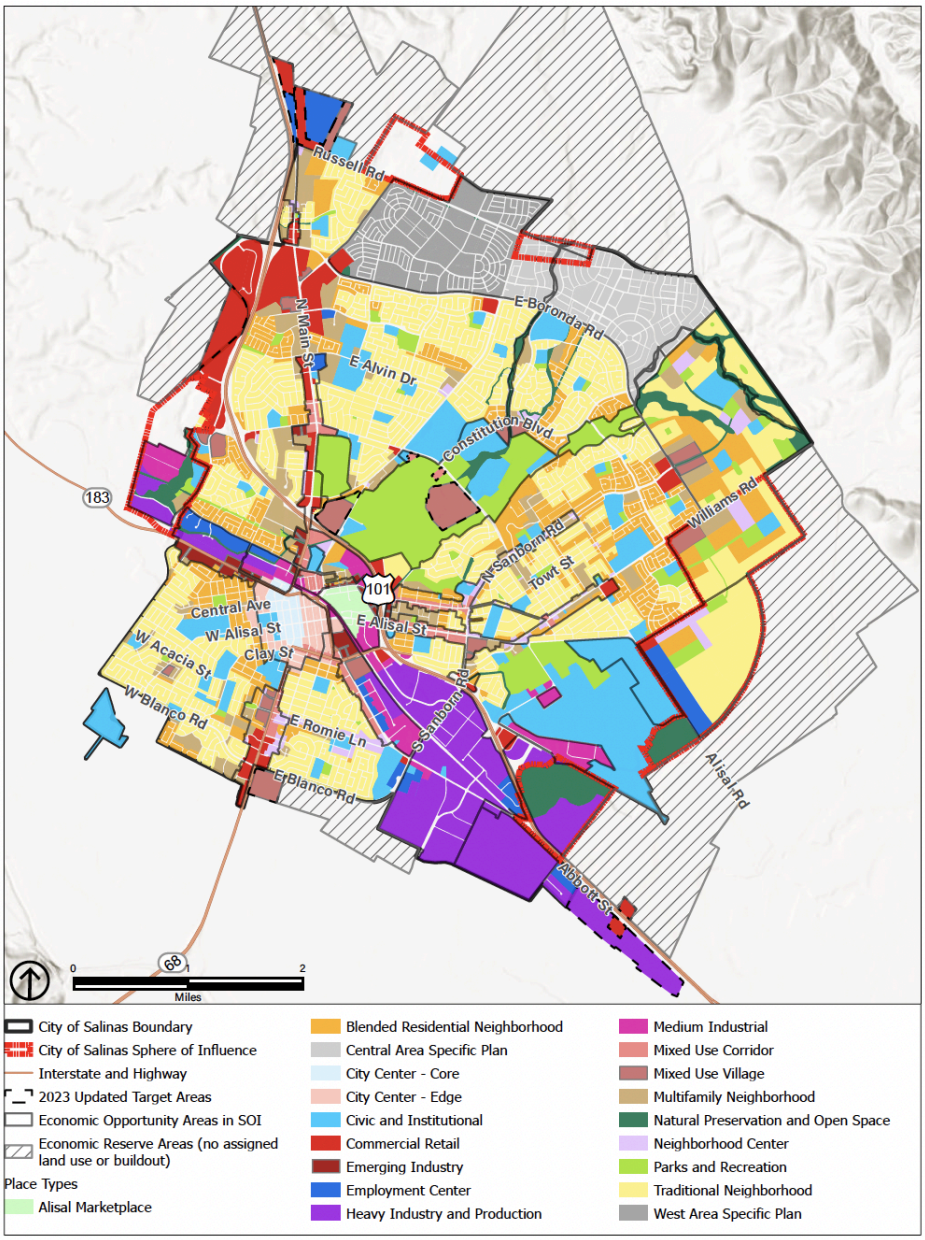

Figure 3.2.1: Land Use Designations for the City of Salinas. There are many interwoven land uses in Salinas. Parks are central, while heavy industry lines Highway 183 as it cuts through the Southern half of the city. Blended neighborhoods, mixed-use corridors, and multifamily neighborhoods are scattered throughout the municipality. Source: City of Salinas General Plan Land Use Map

3.3 City of Soledad

Soledad's land use planning operates within a clearly defined growth boundary framework established to balance development with agricultural preservation. The City is guided by its 2005 General Plan, currently undergoing an update, and three key geographic boundaries: City Limits (2,995 acres), a Sphere of Influence (710 acres), and an Urban Growth Boundary (UGB) of 4,084 acres. The UGB, formalized in 2016 through an agreement with Monterey County,

explicitly constrains urban expansion to protect surrounding agricultural lands from incremental sprawl.

Two major Specific Plans structure detailed development within Soledad's planning area. The Downtown Specific Plan (2012) focuses on revitalization of the Downtown Core, establishing a distinctive development code with zones (Downtown Core, Downtown Edge, and Gateway) designed to promote pedestrian-oriented commercial and civic activities. The Miramonte Specific Plan (2018) represents a more substantial development initiative, planned for 647 acres of former agricultural land and proposing 2,392 residential units (including affordable, medium, and senior housing), neighborhood commercial space, two elementary schools, parks, and public facilities. Miramonte is explicitly designed to support local jobs-to-housing balance and represents a significant commitment to planned growth within the UGB.

Soledad's zoning ordinance establishes residential districts ranging from Single Family (R-1, 1–6 units/acre) through High Density (R-3, 13+ units/acre), along with commercial and industrial designations, as shown in the City of Soledad Land Use Map displayed in **Figure 3.3.1**. Notably, agricultural zoning comprises only 1.4% of land within city limits, reflecting the City's commitment to limiting agricultural encroachment within incorporated boundaries. The 2005 General Plan represents the controlling land use document, though the ongoing General Plan update will establish updated assumptions for long-term growth. The UGB and Sphere of Influence were last updated in 2016 and 2022, respectively, establishing firm boundaries for growth through the foreseeable future. This defined growth boundary provides certainty for water planning by constraining the geographic area for which urban water infrastructure must be provided.

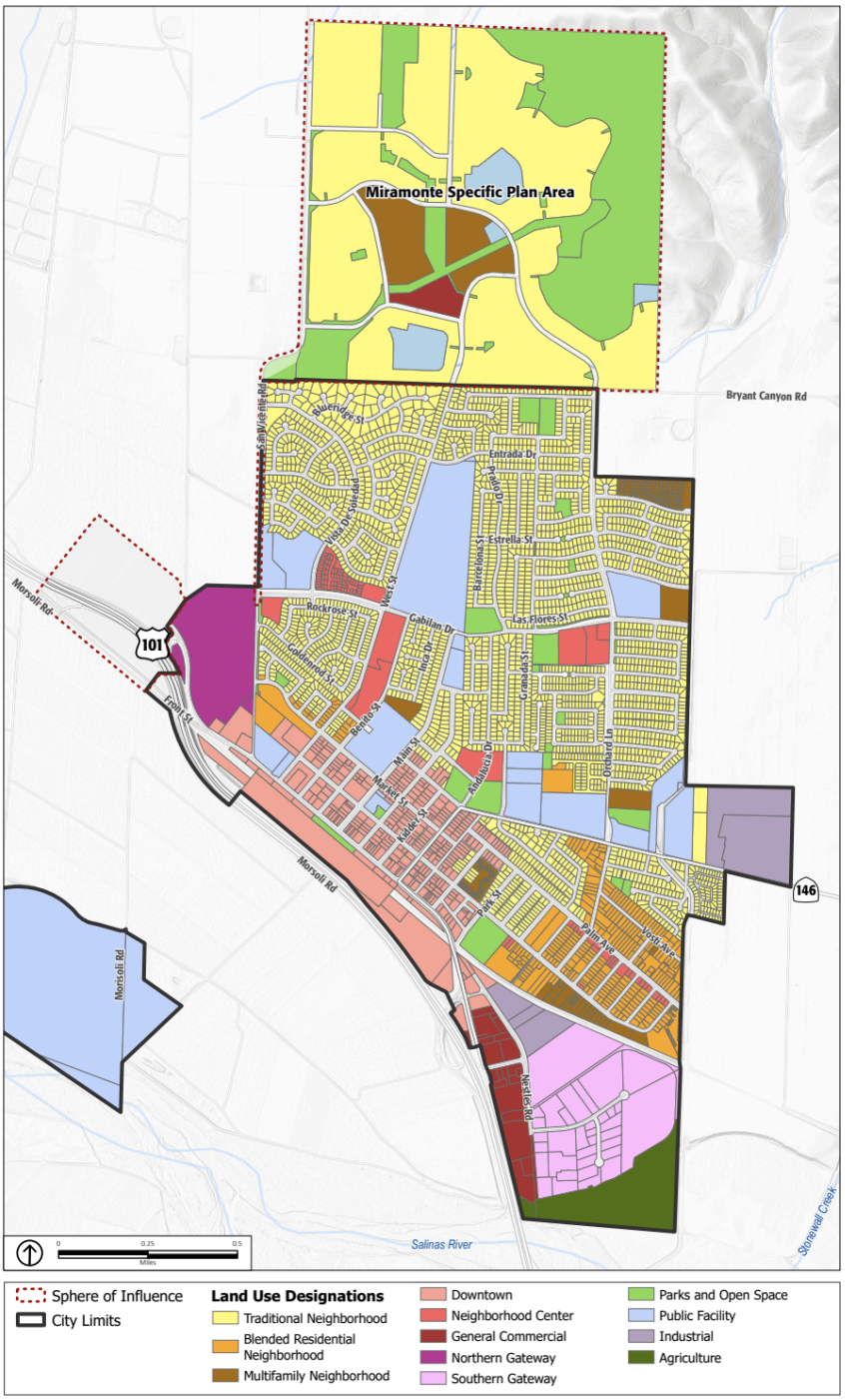

Figure 3.3.1: Land Use Designations in the City of Soledad. There is a minimal mixed land use in Soledad, with sections of the city partitioned off mainly for residential neighborhoods, downtowns and small neighborhood centers. Parks and open space are mainly found in the northern portion of the city as the valley meets the foothills. Source: Soledad General Plan 2045 Draft Land Use Map.

3.4 City of Gonzales

Gonzales operates under the comprehensive Gonzales 2010 General Plan (adopted January 2011, revised June 2018), which explicitly establishes growth boundaries and designations to manage near-term and long-term development while preserving agricultural lands. The plan defines an Urban Growth Area of 2,150 acres available for near-term urbanization and an Urban Reserve Overlay of 2,130 acres intended for long-term growth, with buildout projected around 2050. This dual-boundary approach provides a structured framework for phased infrastructure development and water supply planning over a 40-year horizon.

Gonzales' land use framework includes 19 distinct designations organized around neighborhood-based growth principles, as depicted in the City of Gonzales Land Use Diagram in **Figure 3.4.1**. The Neighborhood Residential (NR) designation, covering 1,075 acres in the new growth area, requires residential development organized into complete neighborhoods with a minimum average density of 7–9 dwelling units per gross residential acre and strict housing mix requirements ("affordability by design"). The Downtown Mixed Use designation (18 acres) encourages high-intensity, multi-story development (0.5–3.0 FAR) with residential densities up to 24 DU/Acre to maintain the historic center as a civic and cultural focal point. A new Community Commercial Mixed Use designation (90 acres) east of Highway 101 is designed around a "Village Center/Main Street" concept and requires Specific Plan approval. Industrial and manufacturing designations (423 acres) are located west of Highway 101 and at interchanges to support job growth and agricultural services.

Interestingly, Gonzales' planning approach explicitly prioritizes agricultural land preservation. The Urban Growth Boundary is bounded by a Permanent Agricultural Edge Overlay, ensuring permanent protection of neighboring farmlands. The plan directs future urban growth eastward toward less productive soils, preserving high-quality agricultural lands west of Highway 101. This strategic orientation reflects watershed-level concerns about maintaining agricultural viability and associated irrigation demands.

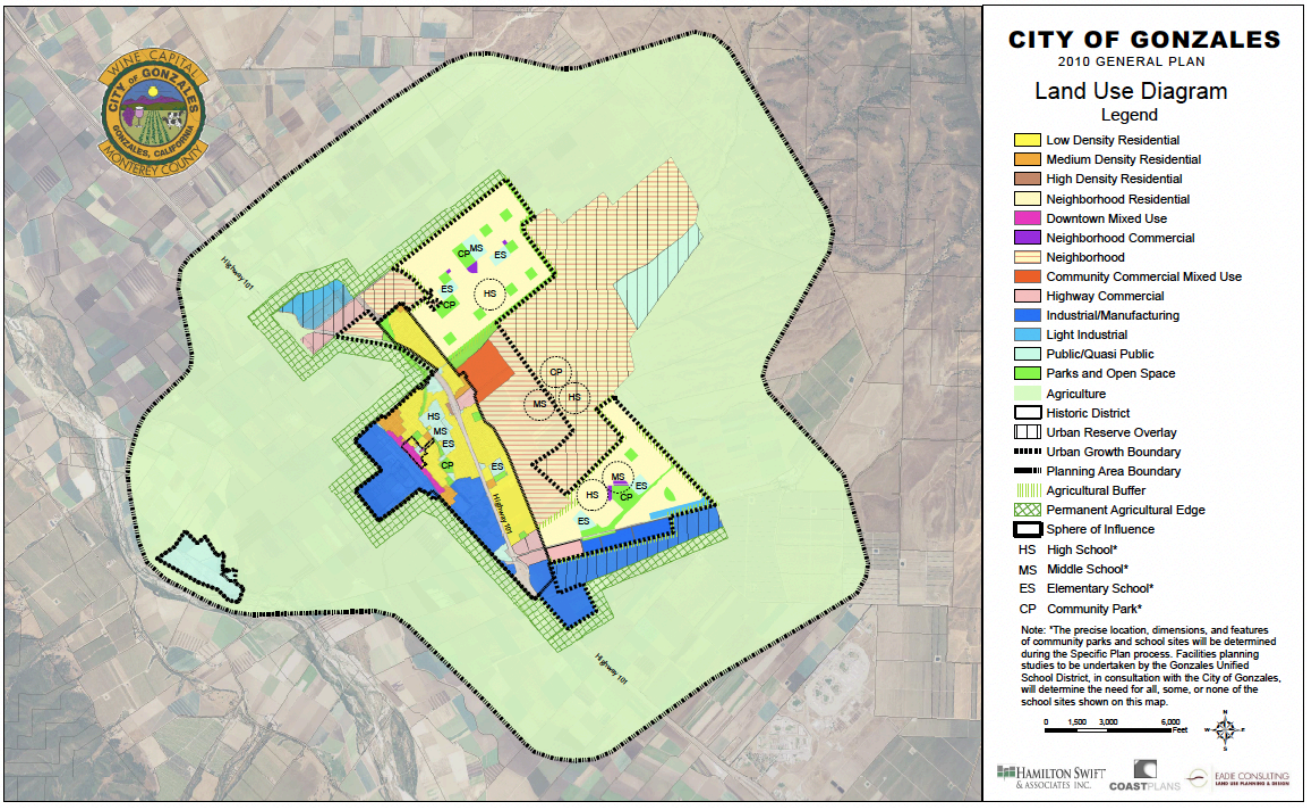

Figure 3.4.1: Land Use Designations for the City of Gonzales. Gonzales consists mostly of low density residential areas and neighborhoods. There are a couple pockets of industry and manufacturing on the outer edges with agricultural land surrounding the municipality. Compared with Salinas and Soledad, Gonzales has considerably lower diversity of land use types. Source: City of Gonzales 2010 General Plan Land Use Diagram.

3.5 Water Management Implications

Land use planning across the three primary urban centers in the Salinas Watershed reflects a coordinated "City-Centered Growth" approach that channels development to defined urban areas while maintaining buffers for agricultural preservation. Salinas' two major specific plans will accommodate approximately 8,200 new housing units by 2045; Soledad's Miramonte Specific Plan adds 2,392 units within a defined UGB; and Gonzales projects buildout around 2050 with phased development within established growth boundaries. These growth projections create quantifiable water demand scenarios that must be coordinated with available surface and groundwater supplies through each City's UWMP and with regional water management. Concurrent with growth management, all three cities employ Agricultural Overlay designations and/or Permanent Agricultural Edge policies to protect high-quality farmlands, recognizing that agricultural water demand remains a dominant factor in watershed hydrology and that preservation of productive agricultural land supports regional economic stability. Effective implementation of these land use plans depends on timely infrastructure development, reliable water supply, and adaptive management as conditions and growth rates evolve.

3.6 Flood Control Zoning and Development Restrictions

Flood hazard management represents a critical overlay on land use planning throughout the Salinas Watershed, particularly in flood-prone areas along the Salinas River, Arroyo Seco River, and tributary channels. Monterey County policy directs that FEMA 100-year flood areas remain free from future development. At the city level, Gonzales implements specific restrictions on development in flood zones, requiring annual identification and review of flood-prone areas, including lands along the Salinas River, Gonzales Slough, and Johnson Canyon Creek. Development within the 100-year flood zone of Gonzales Slough is generally prohibited unless mitigation measures ensure no increase in downstream flood hazard. When development occurs in flood-prone areas, structures must either be elevated above the 100-year flood level or incorporate extensive drainage improvements to remove the site from the floodplain.

Both Monterey County and Gonzales emphasize the preservation of the main channels of the Arroyo Seco and Salinas Rivers, which cannot be encroached upon due to their necessity for flood flow capacity, groundwater recharge, and maintenance of riparian habitats. Gonzales has adopted a comprehensive stormwater management approach that incorporates Low Impact Development (LID) principles and BMPs, with the goal of retaining runoff from the 10-year storm event and detaining flows from the 100-year storm. Open space requirements, particularly along Gonzales Slough and Johnson Canyon Creek, serve dual purposes of flood protection and recreation, with stormwater detention areas jointly used for public access where feasible, provided structures remain outside the 100-year flood zone.

3.7 Specific Designated and Beneficial Uses

The water bodies within the Salinas Watershed support multiple designated beneficial uses that must be protected and coordinated through land use planning and water management. The primary water bodies — including the Salinas River, Arroyo Seco River, Gonzales Slough, and the underlying groundwater aquifers — serve as a drinking water supply, an agricultural irrigation source, an industrial supply, a habitat, and a recreation area. A full breakdown of the designated and beneficial uses can be found in **Table A1** of the *Appendix*.

Drinking Water Supply (Municipal and Domestic)

Groundwater serves as the sole source of potable water supply for Gonzales and the City of Salinas (served by Cal Water). All groundwaters in the Central Coast Basin are generally considered suitable as municipal supply (MUN) where Total Dissolved Solids (TDS) remain below 3,000 mg/L. Surface water is not available as a potable water supply source in the Gonzales area, making groundwater the critical infrastructure for both municipal and domestic use. This designation highlights the interdependence of land use planning and groundwater management: urban growth cannot proceed without demonstrating sustainable groundwater availability and protecting water quality from contamination.

Agricultural Water Supply

Extensive agricultural land surrounding the three cities, including approximately 1,633 acres of prime farmland within Gonzales' urban growth boundary, relies on groundwater extraction for irrigation. As urbanization progresses, agricultural wells will be retired and converted for urban use. Natural surface water features, including Johnson Canyon Creek and McCoy Creek, have been channelized into agricultural ditches and retention basins to divert seasonal runoff and store irrigation water for agricultural use. These engineered surfaces remain essential to current agricultural operations and represent potential alternative water sources (if restored to natural conditions) for recharge and habitat purposes.

Industrial Water Supply

Groundwater is used for industrial purposes in Gonzales, primarily to support agricultural activities within industrial zones. Gonzales and Cal Water anticipate that recycled wastewater will be a future industrial supply source, with tertiary recycled water intended to meet non-potable demand and reduce potable groundwater extraction for extensive landscape irrigation and industrial processes.

Habitat and Ecosystem Protection

The Salinas River segment near the Gonzales wastewater treatment facility carries a federal designation as critical habitat for the South-Central California Coast steelhead (an Evolutionary Significant Unit). This segment also provides habitat for the western pond turtle. The Gonzales Slough, Johnson Canyon Creek, and associated drainages support sensitive habitats, including freshwater marshes, Pacific Willow riparian woodlands, and aquatic habitats. Groundwater-dependent ecosystems (GDEs) in the region depend on adequate groundwater levels and baseflow contributions to surface waters. The preservation of these habitats through adequate streamflow and groundwater levels represents both an environmental mandate and a constraint on groundwater extraction for municipal and agricultural purposes.

Recreation and Open Space

The Gonzales Slough and Johnson Canyon Creek corridors are recognized as recreational open spaces that support passive recreation (walking, biking) and accommodate future trails and recreational amenities. The Gonzales General Plan designates 97 acres for Parks and Open Space, providing both active and passive recreation opportunities. The confluence of the Arroyo Seco and Salinas Rivers is recognized as a regional natural resource with potential for the development of park and recreational facilities.

3.8 Summary

Land use planning in the Salinas Watershed channels approximately 12,500 new housing units to three municipalities through 2045–2050 while protecting agricultural lands through growth boundaries coordinated with water supply planning. Flood hazard management and protection

of designated beneficial uses—including drinking water supply, agricultural irrigation, habitat, and recreation—represent critical constraints on development patterns.

Section 4: Meteorology and Water Quality Monitoring

4.1 Introduction

Meteorological and hydrologic monitoring in the Salinas River watershed provides the foundation for understanding precipitation patterns, temperature cycles, and streamflow behavior across the region. These datasets support the interpretation of storm timing, runoff generation, groundwater recharge, and broader watershed processes that influence water availability and water quality.

4.2 Meteorology and Flow

Meteorological monitoring in the Salinas River watershed relies on two NOAA Climate Observation Stations located near the City of Salinas. Their positions are shown in **Figure 4.2.1**, and their proximity to the largest urban center in the watershed reflects both historical record-keeping convenience and the absence of long-term stations in the upper watershed. While these stations provide highly reliable datasets, their limited spatial distribution presents challenges for characterizing climatic variability across a watershed that spans over 4,156 square miles and contains distinct climatic subregions.

The Salinas Airport station (GHCND:USW00023233) has maintained a continuous record since 1930, and the Salinas Number 2 station (GHCND:USC00047668) collected data from 1958 to 2018. Both stations provide daily precipitation, maximum temperature, and minimum temperature, making them essential for long-term climate analyses. The temporal resolution and length of record support the identification of seasonal cycles, long-term climatic shifts, and hydrologically significant storm events. However, because both stations are located in the lower watershed, climatic gradients driven by elevation, coastal influence, and inland heating are not directly captured.

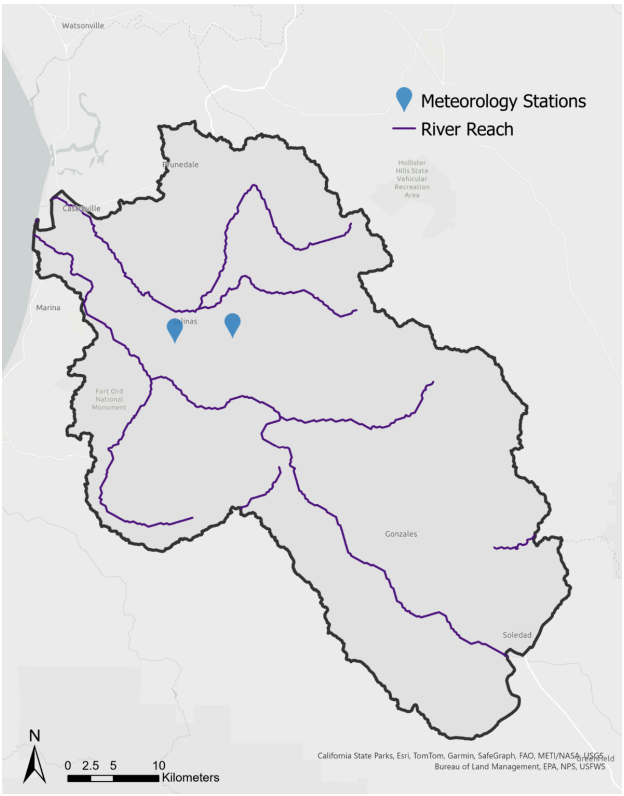

Figure 4.2.1. Two Meteorology Stations in the Salinas River Watershed. Blue markers indicate meteorology station locations while the purple shows river reach within the watershed for context. Both stations are located around the largest urban area—the city of Salinas.

Daily precipitation patterns shown in **Figure 4.2.2** highlight the pronounced seasonality characteristic of Mediterranean climates, with most rainfall occurring between November and March. Interannual variability is evident, with several years exhibiting intense storm clusters followed by extended dry conditions. These oscillations align with broader climatic drivers such as El Niño and La Niña cycles, which are known to influence storm frequency and intensity across coastal California. Years with higher precipitation totals often display multiple high-intensity storm events that contribute disproportionately to annual runoff volumes. Conversely, dry years exhibit sparse rainfall, extended zero-precipitation periods, and limited opportunities for groundwater recharge.

Daily maximum and minimum temperatures presented in **Figure 4.2.3** reveal consistent seasonal oscillations across the monitoring period. Maximum temperatures peak during late summer, while minimum temperatures decline in winter, influencing evapotranspiration rates, irrigation demand, snow presence in upper elevations, and soil moisture dynamics. The similarity between the two stations reflects their close geographic proximity but does not account for colder upland conditions or warmer inland agricultural areas, which may experience temperature extremes not captured in the existing climate record.

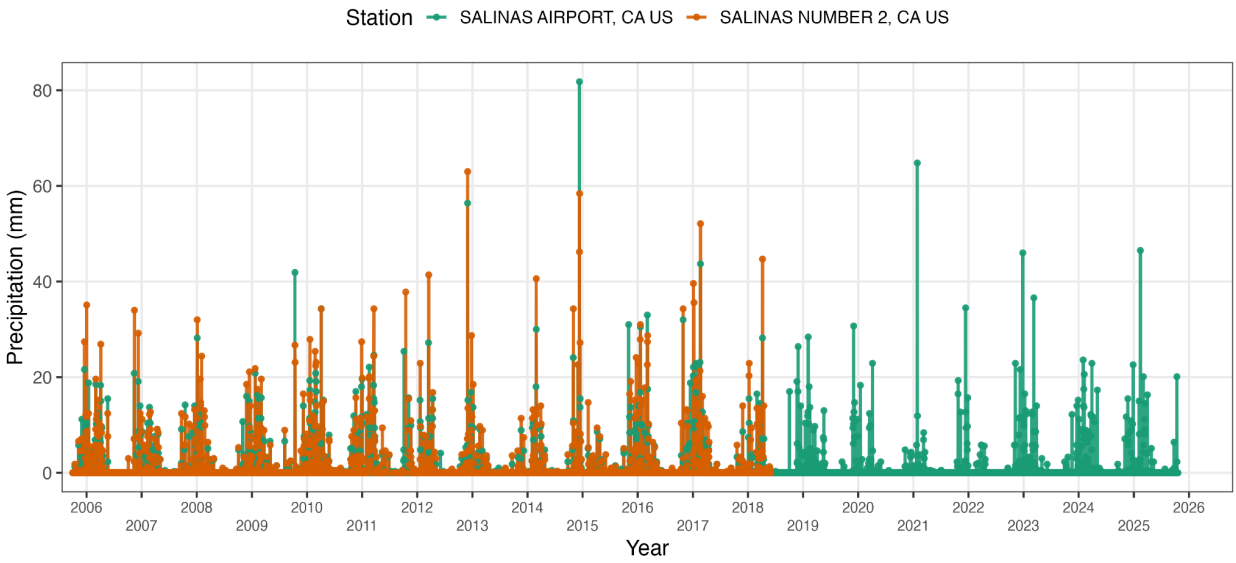

Figure 4.2.2. Daily precipitation recorded at the Salinas Airport and Salinas Number 2 stations. Yearly tick marks allow clear visualization of rainfall patterns, storm timing, and interannual variability, illustrating differences in precipitation behavior across the Salinas Valley.

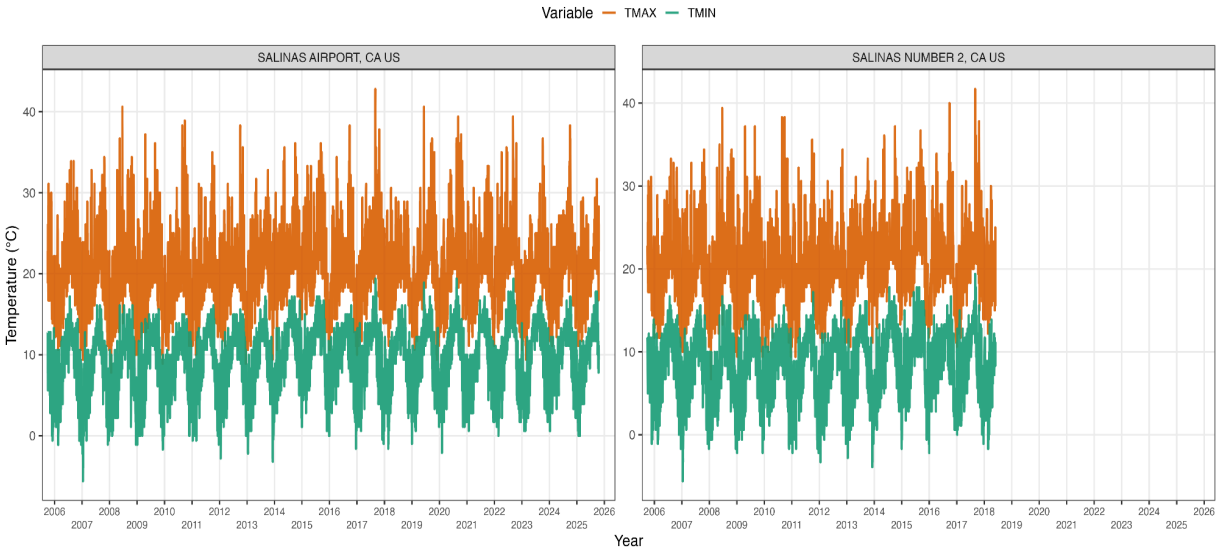

Figure 4.2.3. Daily maximum (TMAX) and minimum (TMIN) temperatures at the Salinas Airport and Salinas Number 2 stations. Yearly x-axis spacing highlights seasonal temperature cycles and longer-term trends, with separate panels providing a clear comparison of conditions between the two monitoring sites.

Hydrologic response to precipitation is measured at two USGS gaging stations located on the main stem of the Salinas River, as shown in **Figure 4.2.4**. The Spreckels (11152500) and Chualar (11152300) stations provide over thirty years of continuous streamflow records. Their placement allows analysis of mid-watershed hydrologic behavior but leaves downstream

dynamics, including potential development areas near Gonzales and Soledad, without direct long-term flow monitoring.

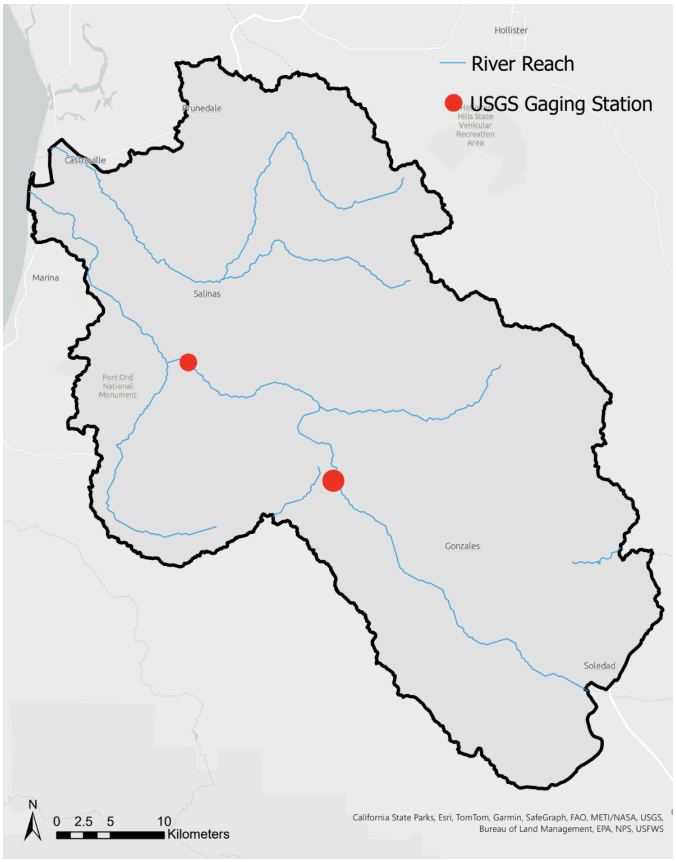

Figure 4.2.4: Location of USGS Gaging Stations within the Salinas watershed. The two USGS gaging stations located in the Salinas river watershed are both located along the main neck of the river. They are strategically located to measure and collect flow data.

Table 4.2.3. USGS gaging stations reporting flow data in the Salinas watershed.

Site	Site Number	Latitude	Longitude	Record Start	Record End
Salinas River near Spreckels, CA	11152500	36.6311N	-121.6714W	01/25/1989	11/06/2025
Salinas River near Chualar, CA	11152300	36.5536N	-121.5483W	10/01/1988	11/06/2025

Storm response patterns shown in **Figure 4.2.5** demonstrate the rapid hydrologic reaction of the watershed to short-duration rainfall events. The hydrograph peaks occur shortly after precipitation pulses recorded at the NOAA stations, indicating flashy runoff behavior influenced by steep upland terrain, compacted agricultural soils, and limited

floodplain connectivity. Large peak flows exceeding 20,000 cubic feet per second at both gages reveal the capacity of major storm systems to mobilize sediment, reshape channel features, and deliver large pollutant loads from tributaries and agricultural sources. The short duration of these peaks reflects rapid recession, suggesting limited surface storage and high infiltration losses once storm energy diminishes.

The combination of long dry periods, sharp storm responses, and frequent zero-flow conditions has implications for both surface water and groundwater systems. Winter storms provide the primary opportunity for groundwater recharge, particularly in permeable alluvial zones such as the Eastern Salinas Valley. However, the limited spatial coverage of meteorological and streamflow stations constrains the ability to quantify recharge contributions from upland tributaries, which may supply significant subsurface flows during intense storm events. Likewise, the lack of monitoring in the southeastern agricultural region limits understanding of stormwater return flows that contribute to both recharge and pollutant mobilization.

Table 4.2.4. Flow measurements for USGS gaging stations in the Salinas watershed.

Site	Site Number	Mean Daily Flow	Minimum Daily Flow	Maximum Daily Flow
Salinas River near Spreckels, CA	11152500	190.73 cfs	0 cfs	20,300 cfs
Salinas River near Chualar, CA	11152300	233.84 cfs	0 cfs	21,400 cfs

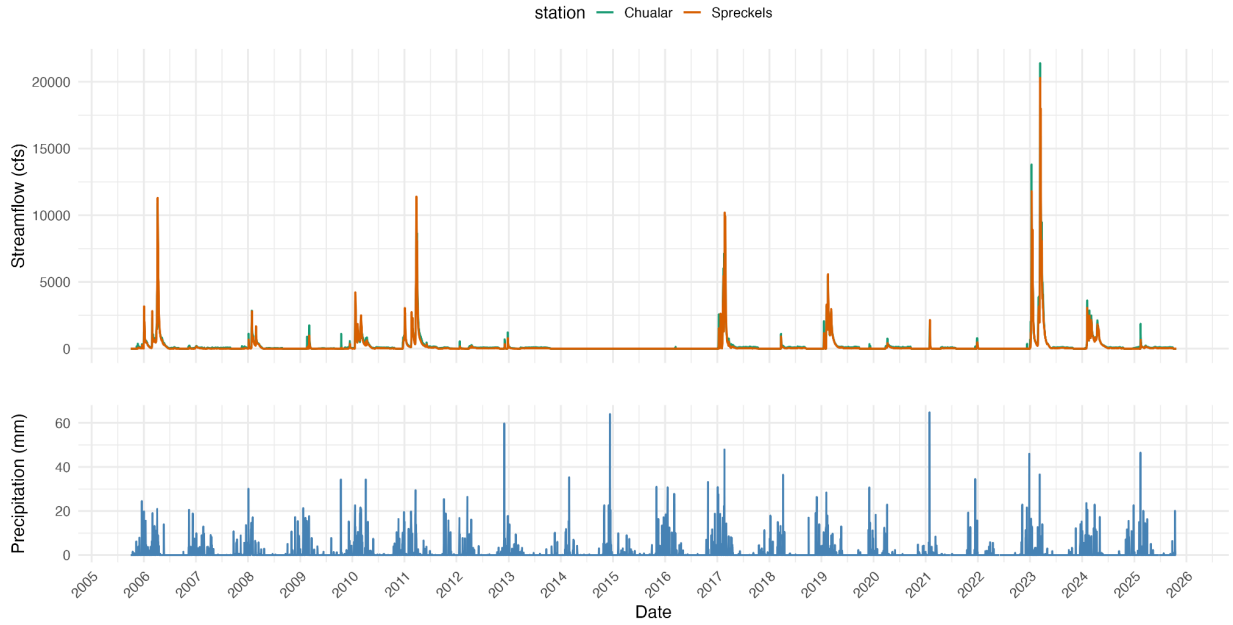

Figure 4.2.5. Daily streamflow (top) and daily precipitation (bottom) for the Salinas River watershed. The precipitation series highlights the timing of storm events, while the corresponding streamflow hydrograph shows how each storm produces a runoff response at the Spreckels and Chualar USGS gauges.

Overall, the meteorological and flow data highlight a watershed characterized by highly variable precipitation, rapid hydrologic responses, and strong seasonal contrasts in water availability. These patterns underscore the importance of integrating climate records, streamflow observations, and watershed characteristics to support accurate modeling of runoff, recharge, flood behavior, and water quality dynamics across the Salinas River system.

4.3 Water Quality

Water quality monitoring in the Salinas River watershed reflects more than two decades of coordinated efforts to address nutrient pollution, sediment impairment, salinity stress, and the impacts of agricultural runoff. The distribution of water quality stations shown in **Figure 4.3.1** highlights that sampling locations are densely clustered along the mainstem and in tributaries that drain agricultural areas between Soledad, Gonzales, Salinas, and Prunedale. This spatial clustering aligns with known impairment zones identified in regional TMDLs and Central Coast water quality assessments and illustrates where monitoring programs have historically focused their attention. Monitoring across the watershed is carried out by multiple programs, including the Central Coast Ambient Monitoring Program, the Cooperative Monitoring Program for irrigated agriculture, municipal stormwater programs, and the USGS Groundwater Monitoring Network.

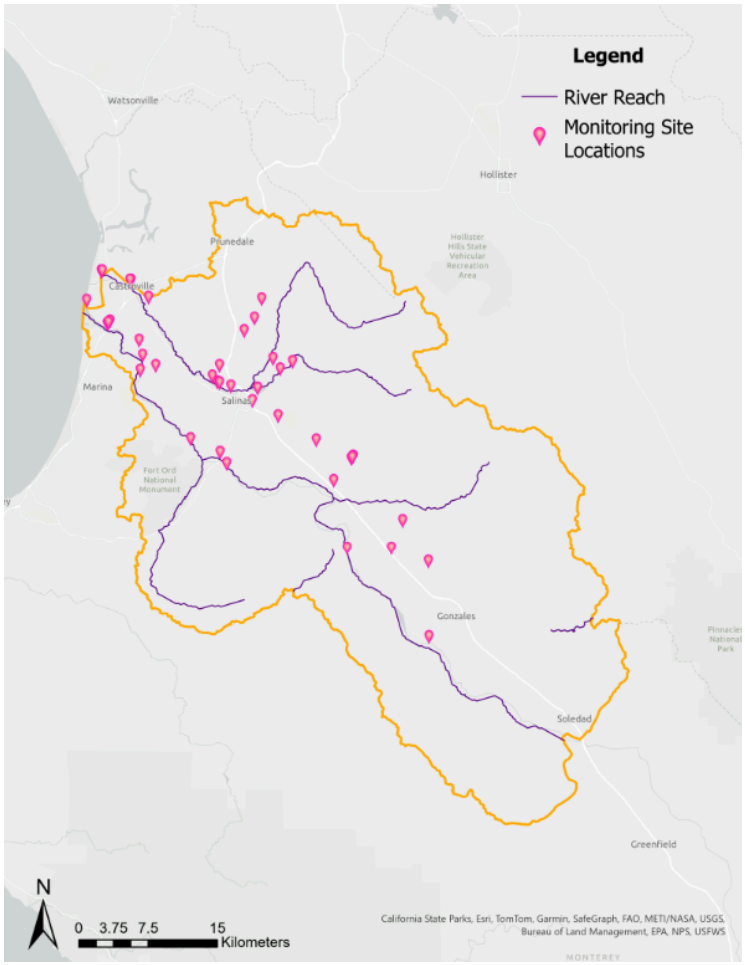

Figure 4.3.1. Water quality monitoring sites within the Salinas River Watershed. Pink markers indicate monitoring site locations throughout the watershed while purple lines represent river reach. Source: National Water Quality Monitoring Council.

Together, these monitoring efforts generate datasets that capture nutrients, sediment indicators, trace metals, pesticides, salinity measures, and general water quality parameters. **Table 4.3.1** summarizes the parameters monitored at individual station locations within the National Water Quality Monitoring Council dataset and illustrates the breadth of analytes tracked across the watershed.

Table 4.3.1. NMQMC water quality monitoring stations and monitored parameters in the Salinas watershed.

Station	Water Quality Parameters
CEDEN – Alisal Creek @ Hartnell Rd dogleg	Ammonium
CEDEN – Asilomar State Park at bridge	Copper
CEDEN – Carmel River Beach mussels	Oxygen
CEDEN – Carmel Storm Drain at the end of Ocean Avenue	Total Nitrogen
CEDEN – Carneros Creek at Blohm Road	Nitrate, Zinc, Magnesium

CEDEN – Carneros Creek at Johnson Road	Turbidity, Chloride
CEDEN – Carneros Creek at mid entry to wetland	Ammonium
CEDEN – Carneros Creek, Sill Road (road flow)	Silt
CEDEN – Central and 13th Streets in Pacific Grove	Total Suspended Solids
CEDEN – Chualar Creek @ Old Stage Rd (south branch)	Sand
CEDEN – Elkhorn Slough Upper near RR tracks	Moisture content, Hydrogen sulfide
CALWR_WQX – LITTLE SUR R A HWY 1	Chlorophyll a
CEDEN – Majors Creek lower	Imidacloprid
CEDEN – Monterey Harbor, random site 24	Urea
CEDEN – Monterey Harbor, random site 28	Total dissolved solids, Conductivity, Potassium
CEDEN – Monterey storm drain at San Carlos Beach	Nitrate, Zinc, Magnesium
CEDEN – Pacific Grove storm drain between Pico and Arendo	Calcium, Iron
CEDEN – Pebble Beach storm drain @ 17 Mile Dr	Fipronil Desulfinyl
CALWR_WQX – Salinas River at Blanco Dr	pH, Water Temperature
CALWR_WQX – Salinas River at Davis Rd	Oxygen
CALWR_WQX – Salinas River near Spreckels	Aluminum
CALWR_WQX – Salinas River 1.9 miles above Hwy 1 Bridge	Fipronil, Fipronil Sulfide

Statistical patterns in the dataset indicate strong differences in sampling frequency and variability among parameter groups. Nitrogen species dominate the dataset, and **Figure 4.3.2** shows that nitrate, inorganic nitrogen, nitrite, and ammonia have each been measured more than one thousand times. These high counts reflect long-standing regulatory attention and the significance of nitrate contamination in both groundwater and surface water. Variance in nitrate concentrations is substantial, especially during winter storms that mobilize stored soil nitrogen and irrigation return flows. These peaks frequently exceed drinking water benchmarks, and chronic exceedances during low flow periods signal the influence of nitrate rich groundwater discharge entering the channel.

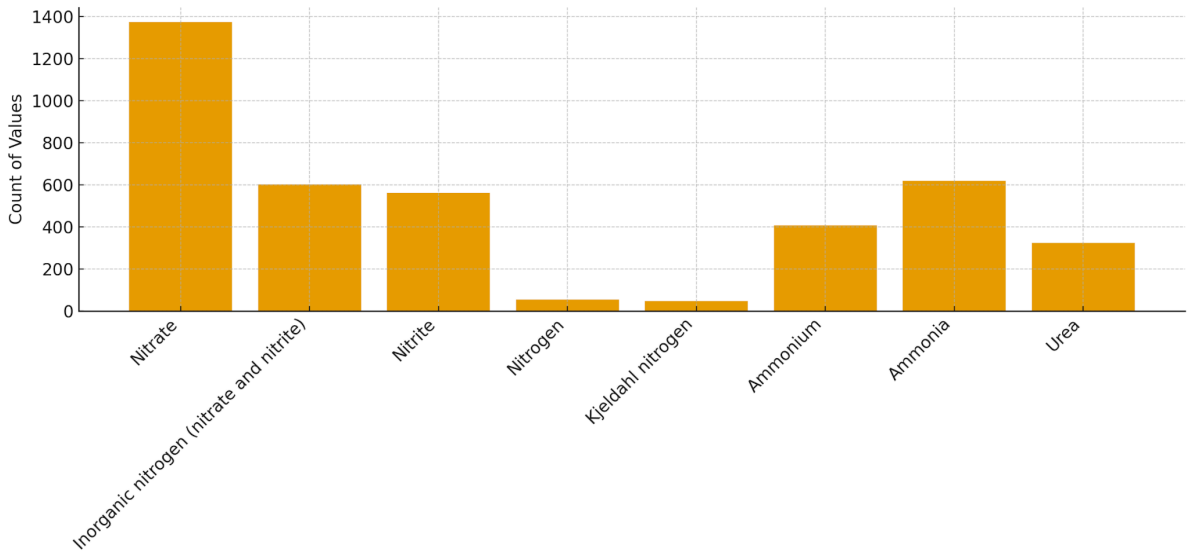

Figure 4.3.2. Count of Monitoring Efforts for Each Species of Nitrogen. Nitrate and inorganic nitrogen dominate the dataset, indicating targeted monitoring for nutrient loading, while other nitrogen forms such as Kjeldahl nitrogen and urea are measured less frequently.

Sediment indicators such as turbidity, suspended sediment concentration, and total suspended solids also show moderate to high sampling frequency and large variability, as illustrated in **Figure 4.3.3.** Storm flows typically generate sharp increases in sediment concentrations, often an order of magnitude higher than dry season baseline conditions. Conductivity and salinity exhibit equally large fluctuations, with elevated values during late summer and early fall that indicate evaporative concentration and potential seawater intrusion in downstream reaches. In contrast, metals and pesticides are monitored far less frequently. Sparse sampling restricts the ability to quantify exceedance frequencies, although available measurements show episodic spikes in contaminants such as copper and diazinon in tributaries like Carneros Creek.

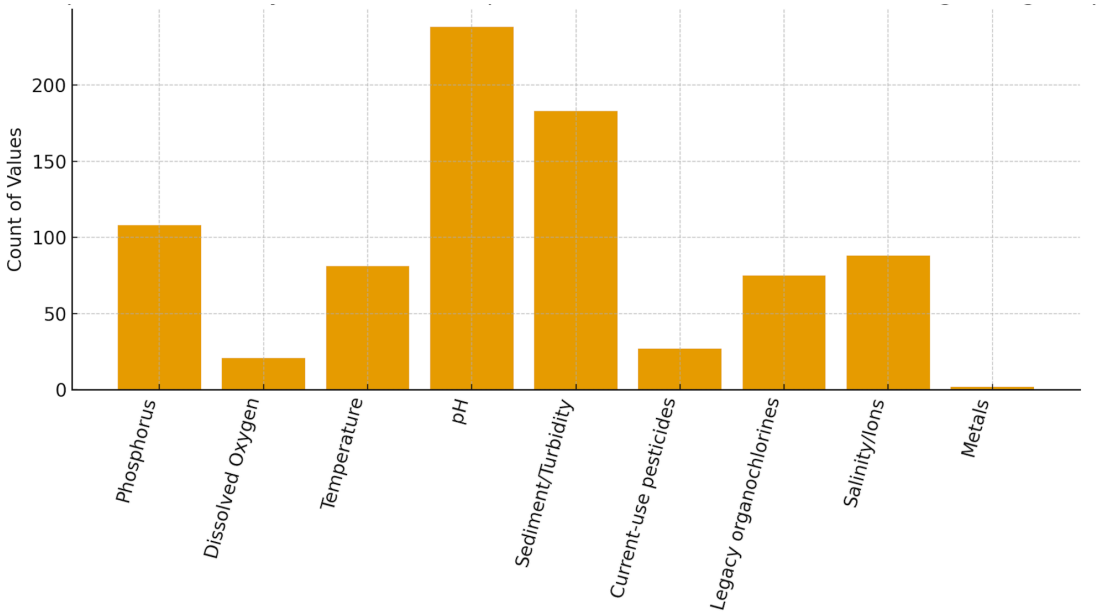

Figure 4.3.3. Count of Monitoring Values for Major Parameter Groups (excluding nitrogen species). Higher counts for parameters such as pH, sediment/turbidity, and salinity/ions indicate greater monitoring coverage, while contaminants such as pesticides and metals have been sampled less frequently.

The observed surface water patterns are closely connected to groundwater pathways across the Salinas Valley. Elevated nitrate during low flow conditions is consistent with known nitrate plumes in shallow alluvial aquifers, where concentrations often exceed drinking water standards. These groundwater surface water interactions are most apparent in agricultural subbasins where shallow groundwater discharges to the channel. Similarly, downstream salinity patterns match documented seawater intrusion fronts that affect both shallow and deep aquifers near Castroville and Marina. High conductivity in surface water near these regions often reflects the upwelling of saline groundwater. Sediment pulses during storms can also transport adsorbed nutrients and metals into riparian aquifers, which highlights the role of infiltration processes in distributing contaminants between surface water and groundwater.

Together, these results highlight several important management considerations. High nitrate concentrations and frequent exceedances underscore the need for improved nutrient management strategies, including optimized fertilizer timing, cover crop adoption, and enhanced irrigation efficiency. For example, nitrate values at long-running stations on Alisal Creek and the lower Salinas River frequently exceed the 10 mg/L drinking water standard, with winter storm pulses driving concentrations several times higher as runoff mobilizes stored soil nitrogen. Similarly, turbidity and total suspended solids measurements at Chualar Creek and Carneros Creek show sharp storm-driven peaks, often reaching levels associated with documented sediment impairment in regional TMDLs.

These pulses coincide with high-energy runoff from fields and eroding tributary channels, indicating that erosion control should prioritize compacted soils, unvegetated drainage ditches, and unstable banks. Salinity indicators follow an equally meaningful pattern. Conductivity and

TDS values at stations near Blanco Road and the lower river consistently rise during late summer and early fall, corresponding with evaporative concentration and the inland migration of the seawater intrusion front. These elevated salinity levels are aligned with GSA-reported groundwater chloride and TDS increases in nearby coastal aquifers. Finally, the infrequent sampling of pesticides and metals reveals clear data gaps for toxic contaminants, even though available measurements such as imidacloprid at Majors Creek or copper spikes in urban storm drains show episodic exceedances of aquatic life criteria. Collectively, these examples indicate that nutrient management, sediment control, salinity mitigation, and targeted contaminant sampling should remain central to watershed protection and restoration strategies.

4.4 Gaps in Monitoring

One potentially problematic trend revealed in the National Water Quality Monitoring Council site map is that the majority of water quality monitoring stations are located in the northwestern, downstream corner of the watershed. While this concentration may be effective for measuring total accumulated pollutants and water quality near end uses in the Salinas urban area and at the river mouth into Monterey Bay, it can lead to limited upriver data and difficulty in attributing pollutant sources to specific subbasins. The southeastern portion of the watershed, particularly around Gonzales and Soledad, which are areas with intensive agricultural activity, lacks adequate flow and water quality monitoring infrastructure. This creates a critical blind spot for understanding pollutant loading and transport dynamics in the upper reaches of the system.

Additionally, the USGS Groundwater Monitoring Network has fallen short of its proposed 317 wells and 8 stream gaging station goals and currently relies on only 214 pre existing wells due to limited funding (CA Water Board). This represents a shortfall of about one third of the planned monitoring capacity. The intended locations for these 103 additional wells and 8 stream gaging stations are not clearly documented, but the original goals were likely designed to provide coverage across the full watershed. The current, stripped back network provides only a minimum level of groundwater monitoring. Given that growers are responsible for more than 90 percent of groundwater withdrawals in the area, the lack of comprehensive groundwater monitoring is particularly concerning for tracking seawater intrusion, aquifer depletion, and nitrate contamination patterns.

Furthermore, the spatial clustering of both meteorological and flow gaging stations near the City of Salinas leaves large portions of the watershed without localized precipitation and discharge data. This limitation makes it difficult to establish accurate hydrologic relationships between storm events and runoff responses across different sub watersheds that have varying land use intensities, soil types, and management practices..

4.5 Summary

Meteorological, streamflow, and water quality monitoring in the Salinas watershed reveal a highly variable precipitation regime with rapid hydrologic responses, persistent nitrate

contamination from agricultural sources, and sediment impairment during storms. At the same time, critical monitoring gaps exist, particularly in the southeastern agricultural region around Gonzales and Soledad and within the groundwater observation network, which limit the ability to track pollutant sources and understand upper watershed dynamics that are essential for management and planning.

Observed exceedances of nitrate and other parameters have direct implications for compliance with applicable water quality criteria, including drinking water standards, Central Coast Basin Plan objectives, and coastal receiving water limits under the California Ocean Plan. These links are especially important where the watershed discharges to Monterey Bay and where wastewater treatment plants or other point sources discharge directly to the ocean. Overall, the current monitoring record suggests that strengthening spatial coverage, improving groundwater and surface water integration, and aligning monitoring design with regulatory criteria will be critical for effective water quality management in the Salinas River watershed.

Section 5: Point Source and Nonpoint Source Pollution

5.1 Introduction

Water quality in the Salinas River Watershed is shaped by both regulated point sources and widespread nonpoint pollution from agricultural and urban activities. While permitted wastewater treatment facilities contribute measurable loads of nutrients, salts, and trace metals, the dominant pollution burden originates from nonpoint sources—primarily intensive agricultural production across the region's 345,000+ acres of cropland, which generates nutrient runoff, sediment, and organic pollution that collectively degrade surface water and threaten aquatic ecosystems. Understanding the relative contributions of point and nonpoint sources is essential for developing targeted management strategies that address the most significant pollutant pathways and protect both water quality and beneficial uses throughout the watershed.

5.2 Point Source Pollution

Point source pollution plays an important but relatively limited role in the overall water quality picture of the Salinas River watershed. Unlike nonpoint sources, which are widespread and difficult to track, point sources are discrete and regulated, typically involving permitted discharges from wastewater treatment plants or industrial facilities. In the Salinas watershed, most attention is focused on agricultural runoff; however, permitted dischargers remain a critical component of watershed management due to their consistent inputs and the availability of monitoring data. Although the number of primary point sources in the region is small, their discharges can still introduce nutrients, salts, and toxic substances into receiving waters, especially in areas where dilution capacity is low or flow is intermittent.

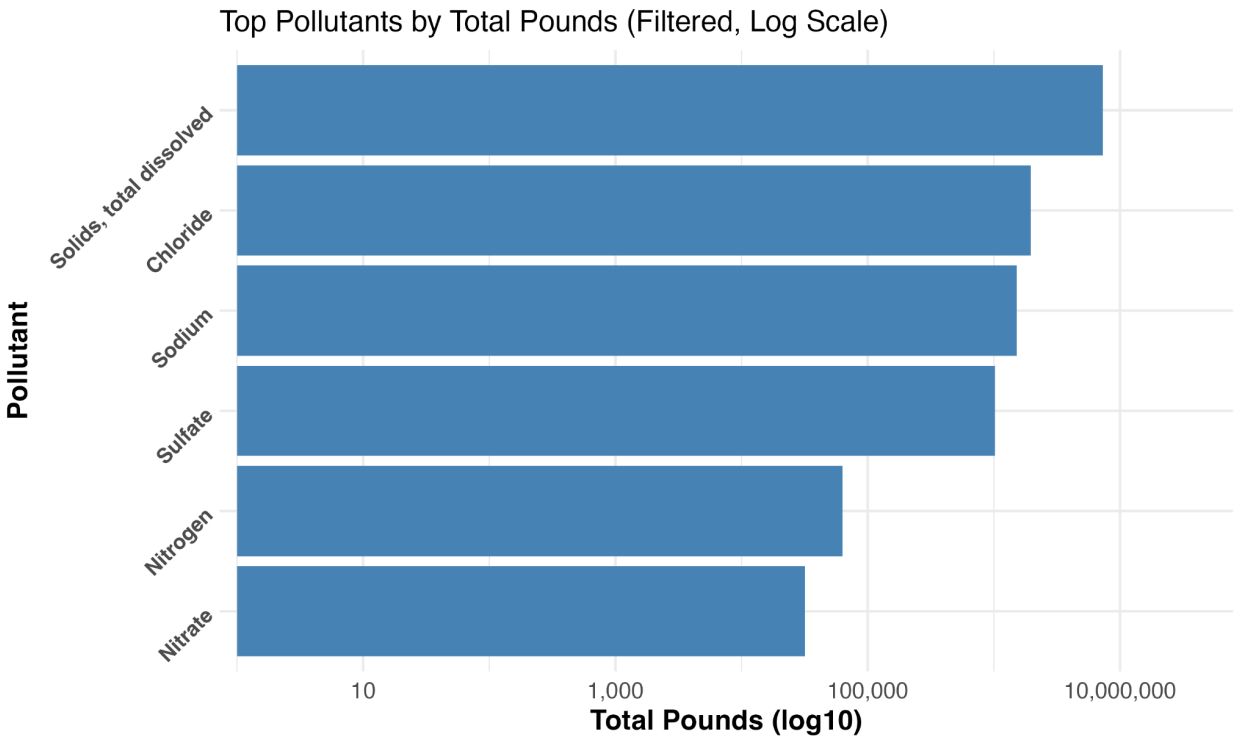

Figure 5.2.1. Top pollutants by total pounds discharged (log scale). This figure shows the pollutants contributing the largest total mass of annual discharge. The y-axis is displayed on a logarithmic scale to account for differences spanning several orders of magnitude. Ammonia, nitrite nitrogen, biochemical oxygen demand (BOD), and total suspended solids were removed because their discharge quantities were small relative to the dominant ions and did not affect interpretation.

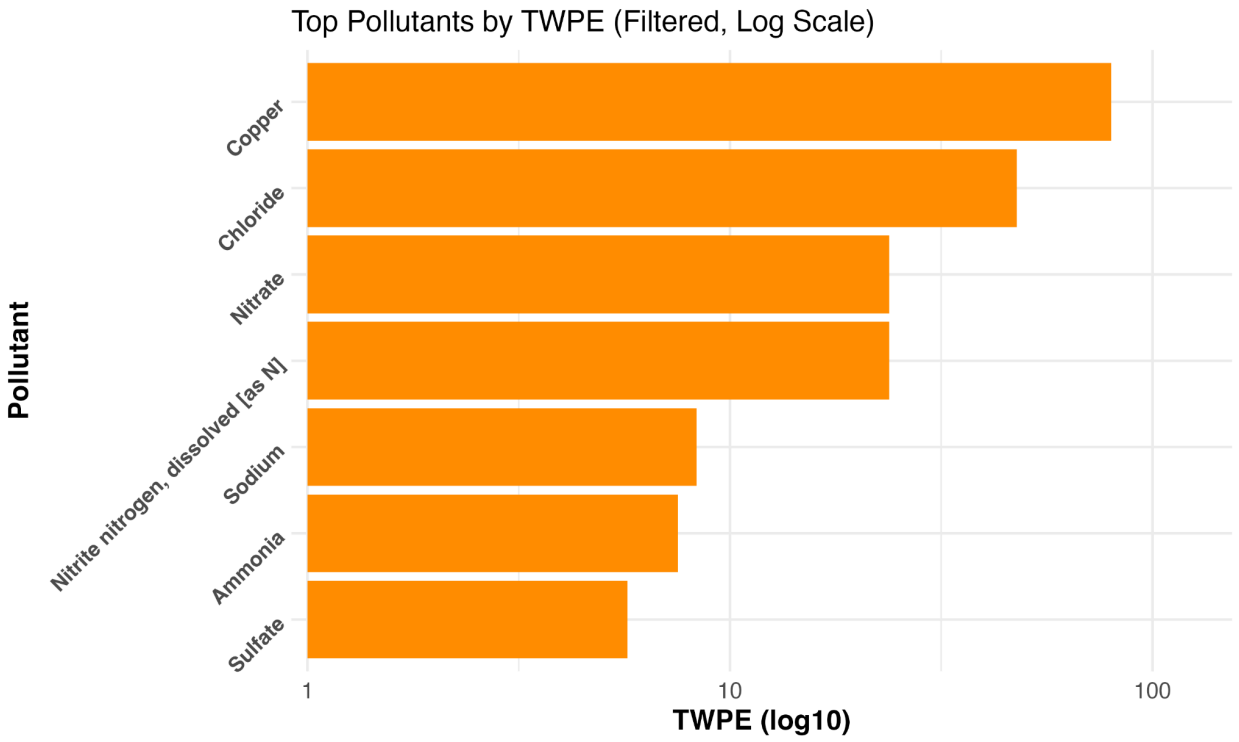

Figure 5.2.2. Top pollutants by Toxic Weighted Pounds Equivalent (TWPE) (log scale). This figure shows the pollutants contributing the highest overall toxicity burden as measured by TWPE. The y-axis uses a logarithmic scale to capture strong differences in toxicity among pollutants. Pesticide compounds alpha-BHC, beta-BHC, and alpha-Endosulfan were removed because their TWPE values were negligible and visually compressed the scale.

Figures 5.2.1 and **5.2.2** summarize pollutant discharges from two permitted wastewater treatment facilities located in the upper Salinas River watershed: the City of Paso Robles Wastewater Treatment Facility and the Heritage Ranch Community Services District. These data were n from the U.S. EPA's ECHO (Enforcement and Compliance History Online) database, which compiles self-reported annual discharge data submitted to the Permit Compliance System (PCS). Figure 1 ranks the top ten pollutants by total pounds discharged per year, with total dissolved solids (TDS), chloride, and sodium emerging as the most dominant by mass. These constituents are commonly associated with domestic wastewater, water softening processes, and legacy salinity inputs. While they may not be acutely toxic, elevated levels of these pollutants can increase stream salinity, impair habitat conditions, and affect agricultural water use downstream.

Figure 5.2.2 presents the same pollutants ranked by their Toxic-Weighted Pounds Equivalent (TWPE), which accounts for both pollutant toxicity and quantity. Here, substances like copper, nitrate, and nitrite rank higher, reflecting their potential to cause ecological harm even in smaller doses. These pollutants are known to affect aquatic organisms and contribute to eutrophication, particularly in systems sensitive to nutrients, such as the Salinas River.

However, a key limitation of the dataset is that no PCS-reported point source data were available for the lower Salinas watershed, where much of the water quality concern and data collection efforts are focused. The two facilities included in this analysis are located entirely in the upper watershed, meaning the results do not capture discharges that may be occurring closer to the estuary or agriculturally intensive zones. This highlights a vital data gap and suggests that while these figures provide insight into regulated discharges, they may underrepresent total pollutant loading to the lower Salinas system, where nonpoint sources likely play a dominant role.

Figure 5.2.3 Aerial Photo of the City of Paso Robles Wastewater Treatment Plant. Source: Treatment Plant Operator (TPO).

The City of Paso Robles Wastewater Treatment Plant as seen in **Figure 5.2.3** serves the urban population of Paso Robles and is a key point source facility in the upper Salinas River watershed. The facility treats an average of 2.9 million gallons of wastewater per day, with a permitted capacity of 4.9 million gallons per day. In 2015, the plant underwent a major upgrade to incorporate Biological Nutrient Removal (BNR), enabling it to better reduce nitrogen and phosphorus levels in compliance with regional water quality goals. Treated wastewater is disinfected, undergoes secondary treatment, and tertiary treated wastewater is either discharged to the Salinas River or reused for irrigation at the nearby city-owned Hunter Ranch Golf Course. The upgraded facility helps protect the watershed by improving effluent quality, but discharge data from the EPA's PCS system show that it still contributes measurable quantities of salts, nutrients, and trace metals to the river. As one of the largest regulated dischargers in the region, the Paso Robles plant plays a significant role in shaping water quality conditions in the upper watershed.

Figure 5.2.4 Heritage Ranch Community Services District's Water Resource Recovery Facility. Source: Project Study Report.

The Heritage Ranch Community Services District's Water Resource Recovery Facility, pictured in **Figure 5.2.4** serves the rural residential community of Heritage Ranch, located near Lake Nacimiento in northern San Luis Obispo County. This facility treats an average of 160,000 gallons per day of domestic wastewater and is designed to serve approximately 2,100 homes. The treatment system includes aerated lagoons, settling ponds, and disinfection, with final effluent disposed of via percolation ponds, allowing treated water to infiltrate into the ground. Unlike the Paso Robles facility, Heritage Ranch does not discharge directly to surface waters under normal conditions, which limits its contribution to surface water pollutant loading. However, the facility still reports discharge data under a National Pollutant Discharge Elimination System (NPDES) permit and may discharge under certain conditions when permitted. Although smaller in scale, the plant plays an important role in protecting local groundwater quality and managing wastewater in an area closely connected to the Nacimiento Reservoir, a key regional water supply.

5.3 Nonpoint Source Pollution

As highlighted in **Table 5.3.2**, Monterey County produces over $5 billion worth of agricultural products in a given year and it is the #1 producer of crops in the state and #1 in the U.S. The predominant crop types by $ volume are vegetables, fruits, berries, and nursery plants, alongside cattle. In 2022, vegetables accounted for 345,176 acres of land use and there were over 50,000 cattle and calves in the county. With this immense production and land use comes immense inputs and nonpoint source pollutants.

Analyzing four of the most significant downstream HUC-12 subregions within the Salinas River watershed allows us to better understand these trends and pollutant loads. As highlighted in **Table 5.3.1**, the primary nutrient load produced within the Salinas River watershed is BOD which indicates the amount of organic pollution and oxygen depletion potential as bacteria break down organic material in the water. This is followed by nitrogen which could come from

organic or inorganic (synthetic) sources such as fertilizers for crops. Lastly, phosphorus load indicates eutrophication potential for watersheds as it is typically the limiting nutrient for freshwater systems. **Figure 5.3.3** allows us to see the primary non-point sources of these runoff pollutants; led by urban, pastureland, and cropland. Combined, cropland and pastureland outweigh urban non-point pollution thanks to agricultural sources such as fertilizer runoff, animal manure, and soil erosion. Urban pollution is likely driven by lawn fertilizer runoff, stormwater runoff on impervious (paved) surfaces, and soil erosion from development projects. For the most part, natural sources do not significantly contribute to non-point pollution, with the exception of forest nitrogen loads.

Table 5.3.1: Total nutrient loads by subwatershed for nitrogen, phosphorus, BOD, and sediment. Source: US EPA PLET.

Watershed	N Load (lbs/year)	P Load (lbs/year)	BOD Load (lbs/year)	Sediment Load (tons/year)
180600051509 - Alisal Creek-Salinas River	116,549	17,223	290,830	3,795
180600110103 - Alisal Slough-Tembladero Slough	64,307	10,899	189,012	2,826
180600110101 - Mud Creek-Gabilan Creek	38,907	3,605	67,599	195
180600110102 - Nativdad Creek-Gabilan Creek	32,286	4,477	96,211	970
Total Load	252,049	36,204	643,652	7,786

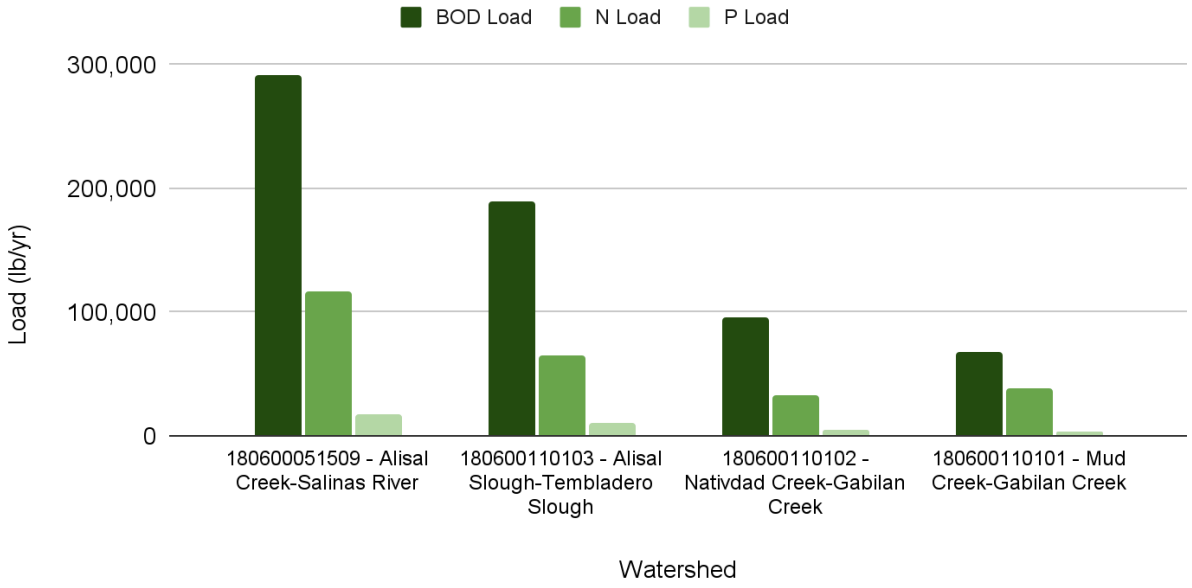

Figure 5.3.1: Total loads for BOD, N, and P in pounds per year by subwatershed. Source: US EPA PLET.

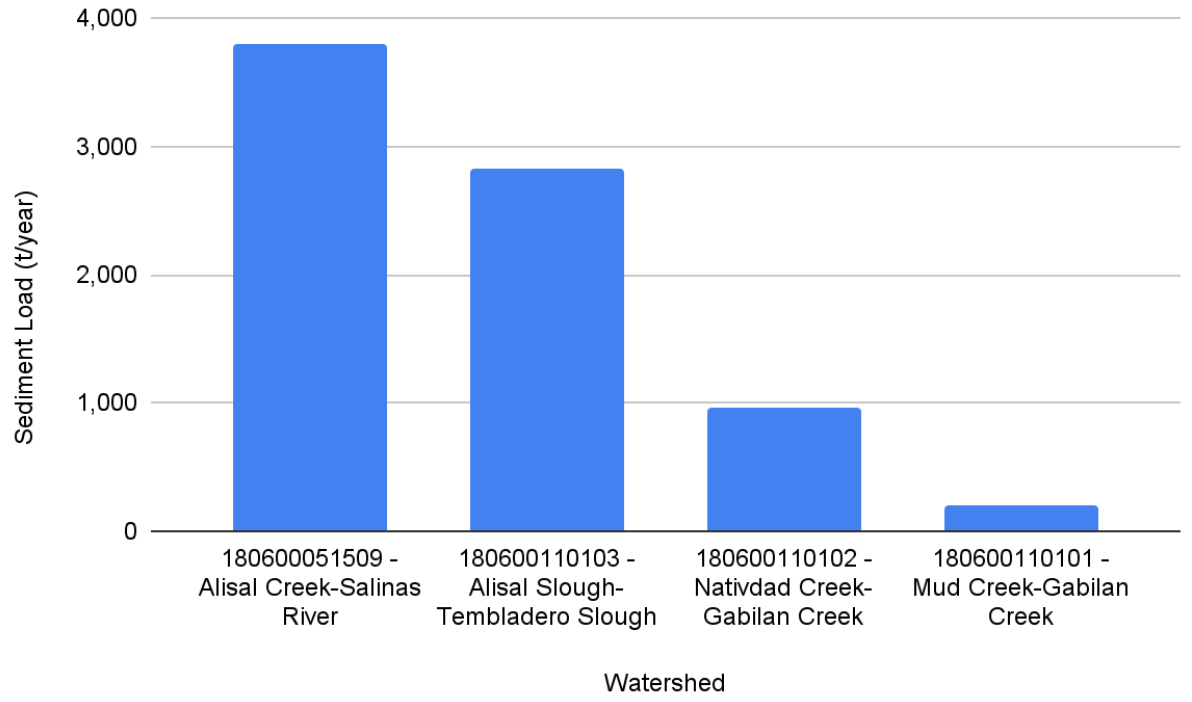

Figure 5.3.2: Total loads for sediment in tons per year by subwatershed. Source: US EPA PLET.

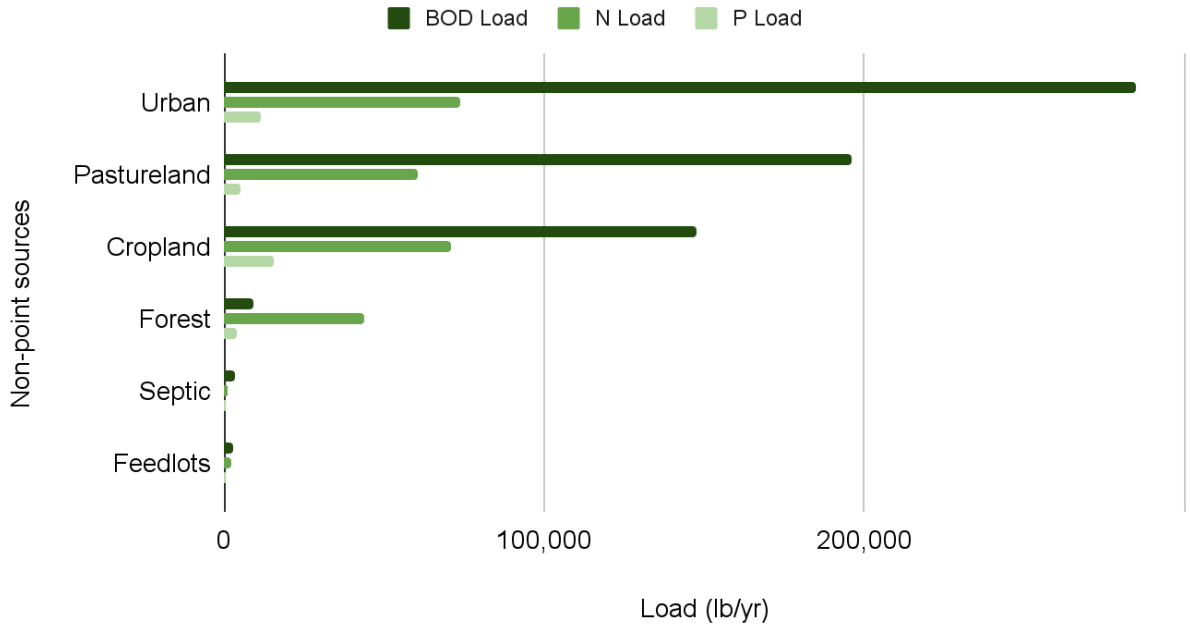

Figure 5.3.3: Total loads by non-point source categories for BOD, N, and P in pounds per year. Source: US EPA PLET.

Table 5.3.2: Total market value of agricultural products sold in Monterey county, 2022. Source: USDA Census.

Market Value of Agricultural Products Sold	Sales ($1,000)	Rank in State [c]	Counties Producing Item	Rank in U.S. [c]	Counties Producing Item
Total	**5,472,081**	**3**	**58**	**3**	**3,078**
Crops	**5,421,074**	**1**	**57**	**1**	**3,074**
Grains, oilseeds, dry beans, dry peas	1,912	26	48	2,040	2,917
Tobacco	-	-	-	-	267
Cotton and cottonseed	-	-	11	-	647
Vegetables, melons, potatoes, sweet potatoes	3,787,636	1	57	1	2,831
Fruits, tree nuts, berries	1,450,153	5	56	5	2,711
Nursery, greenhouse, floriculture, sod	180,069	7	55	19	2,660
Cultivated Christmas trees, short rotation woody crops	50	15	32	391	1,274
Other crops and hay	1,254	36	56	1,862	3,035
Livestock, poultry, and products	**51,007**	**19**	**58**	**1,080**	**3,076**
Poultry and eggs	(D)	15	56	(D)	3,027
Cattle and calves	34,814	13	57	519	3,047
Milk from cows	(D)	23	34	(D)	1,770
Hogs and pigs	(D)	(D)	53	(D)	2,814
Sheep, goats, wool, mohair, milk	282	37	56	695	2,967
Horses, ponies, mules, burros, donkeys	228	38	55	1,043	2,907
Aquaculture	1,151	15	39	167	1,190
Other animals and animal products	58	40	53	1,185	2,909

5.4 Summary

Water quality in the Salinas River Watershed is shaped by both regulated point sources and widespread nonpoint pollution from agricultural and urban activities. While permitted wastewater treatment facilities contribute measurable loads of nutrients, salts, and trace metals, the dominant pollution burden originates from nonpoint sources—primarily intensive agricultural production across the region's 345,000+ acres of cropland, which generates nutrient runoff, sediment, and organic pollution that collectively degrade surface water and threaten aquatic ecosystems. Understanding the relative contributions of point and nonpoint sources is essential for developing targeted management strategies that address the most significant pollutant pathways and protect both water quality and beneficial uses throughout the watershed.

Section 6: Stormwater Analysis and Existing Management Projects

6.1 Introduction

Best Management Practices play an essential role in reducing pollutant loads and improving water quality across the Salinas River Watershed. Because the effectiveness of BMPs depends on local storm characteristics, an urban stormwater analysis is needed to determine the size and capacity that these systems must handle. The Salinas watershed experiences frequent small storms and occasional moderate events, so BMPs must be designed to function during repeated winter rainfall while maintaining adequate storage and treatment capacity.

Storm analysis from PLET model results are used to identify a set of recommended BMPs for cropland, rangeland, and urban areas. These recommendations focus on practices that most effectively reduce sediment, nutrient, and runoff contributions given the watershed’s land-use patterns. Together, the storm analysis and BMP prioritization provide a preliminary framework for managing pollutant sources and improving long-term watershed conditions.

The Salinas Watershed experiences frequent small storms (14-17% of days) but rarely receives large precipitation events, necessitating BMP design that prioritizes management of 2.5 cm storms while maintaining capacity for occasional moderate events. Modeling shows that implementation of riparian buffers, conservation tillage, cover cropping, livestock exclusion, and urban green infrastructure across 25% of agricultural and urban land could achieve 4-21% reductions in pollutant loads, with the greatest improvements in sediment (21%) and phosphorus (18%). Existing regional programs—including reservoir operations, floodplain restoration projects, groundwater sustainability planning, recycled water expansion, and agricultural best management practice compliance—provide a robust foundation for coordinated watershed management through 2045 and beyond.

6.2 Stormwater Analysis

Storm frequency and intensity were evaluated using daily precipitation data from two meteorological stations in the Salinas River Watershed: Salinas 2 (2005–2018) and the more complete Salinas Airport record (2005–2025). Over their respective periods of coverage, the Salinas 2 station recorded **708 storm events**, while the airport station recorded **1,055 storm events** evidenced in **Table 6.2.1**. This corresponds to storm frequencies of **16.7%** and **14.4%**, respectively. As with most Central Coast rainfall patterns, the overwhelming majority of these storms were small: at both stations, nearly all events were less than 2.5 cm of rain. Larger storms were rare, with Salinas Number 2 recording **30 events ≥2.5 cm**, and the airport recording 22 such events, representing just 0.7% and 0.3% of all days with available data, respectively.

Table 6.2.1: Storm Events at the Salinas 2 and Salinas Airport Stations

	Salinas 2	Salinas Airport
Years of Coverage	2005–2018	2005–2025
Total # of Storms	708	1,055
Frequency of Storm Events	16.7%	14.4%
# of Storms <2.5 cm	678	1,033
# of Storms ≥2.5 cm	30	22
Frequency of Storms ≥2.5 cm	0.7%	0.3%

The distribution of storm magnitudes has implications for the design and performance of urban BMPs. Although the Salinas watershed experiences a Mediterranean climate with long dry summers, storms occur frequently enough during the winter months that BMPs must be designed to withstand multiple consecutive wet days. Systems with slow drawdown times—such as retention basins, infiltration trenches, or vegetated swales—may struggle to maintain capacity if not carefully sized to handle repeated storm pulses. Inadequate design could lead to bypass flows or pollutant washout, undermining water-quality objectives.

The storm magnitude–frequency distribution for the airport station displayed in **Figure 6.2.1** illustrates that small storms (0–0.5 cm) dominate the regional rainfall pattern, with steadily declining numbers of events as storm magnitude increases. Nevertheless, storms with 2.5-5 cm of rainfall do occur and represent critical design thresholds for sizing BMPs. At a minimum, BMPs in the Salinas River Watershed should be designed to reliably manage a 2.5 cm (1-inch) storm, since only 22 storms over 20 years exceeded this threshold. Ideally, if budgets and site constraints allow, systems could be built to accommodate 3–5 cm storms, providing resilience against the rare but consequential larger events characteristic of atmospheric–river–driven rainfall.

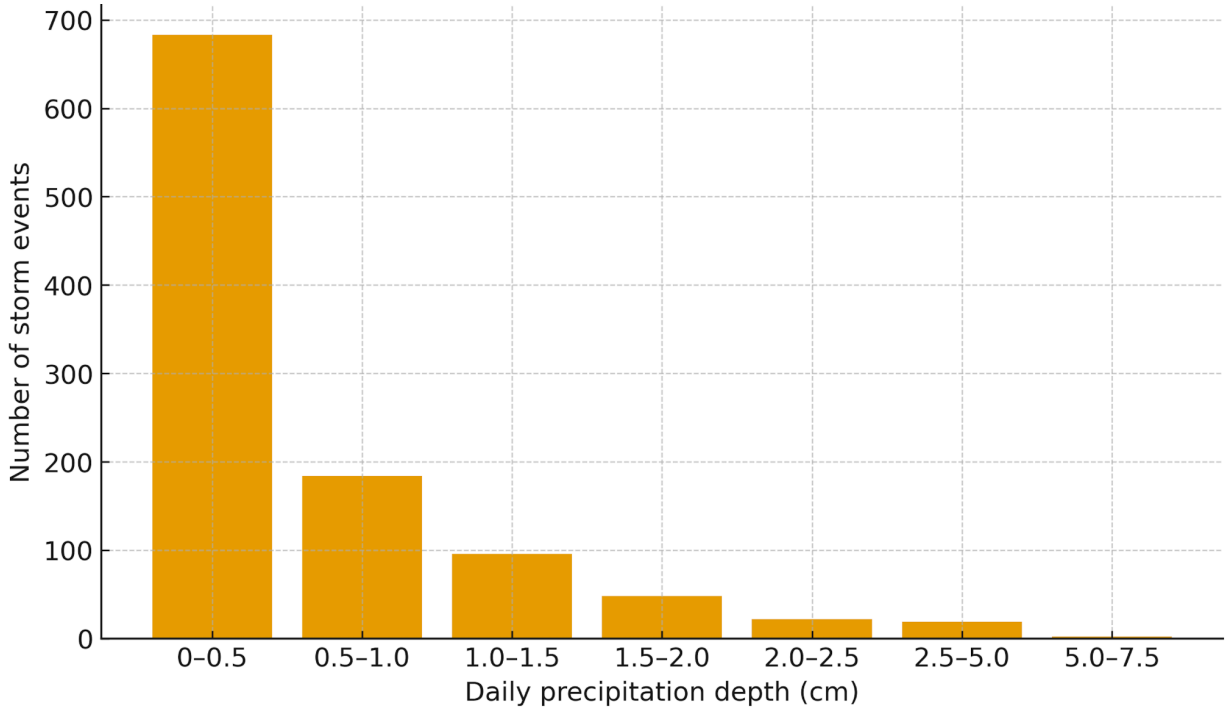

Figure 6.2.1. Distribution of Storm Events by Magnitude at the Salinas Airport Station (2005–2025). The vast majority of precipitation events fall into the smallest category, 0-0.5 cm of daily precipitation. This does not reach the 2.5cm threshold outlined above. Rarely do larger storm events occur, but they are frequent enough to warrant consideration.

Overall, the storm analysis suggests that BMP design in the Salinas watershed should prioritize frequent small storms while maintaining adequate capacity for occasional moderate storms. Properly sizing infrastructure for this rainfall regime is essential to meeting long-term water-quality and flow-management goals.

6.3 BMP Analysis

Due to the fact that 53% of the watershed area is utilized for cropland and pastureland, BMP recommendations are focused on these two sectors (in addition to urban BMPs) and combined multiple BMPs in the EPA's Pollutant Load Estimation Tool (PLET) to establish a more thorough protocol for each land use category. This combination is outlined in the sections below. These treatments were applied to 25% of their respective land-use acreage – assuming the watershed could achieve some improvement but would not be able to overhaul the majority of well-established crop and pasture management practices overnight. The spatial land use patterns are displayed again in **Figure 6.3.1**

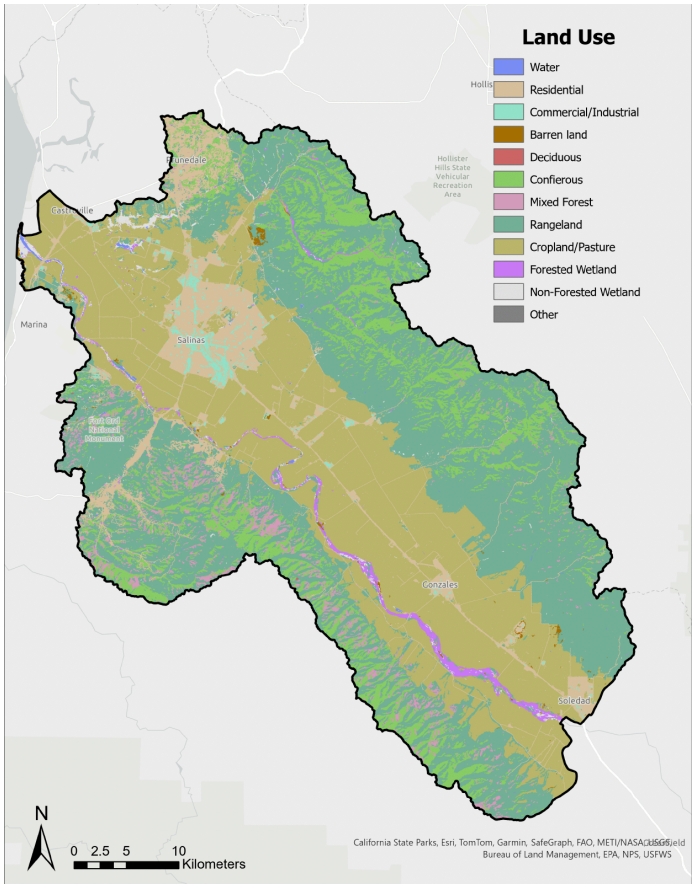

Figure 6.3.1: Multiple Land Use Designations for Salinas Watershed. Upland areas on both sides of the watershed are dominated by coniferous forest and rangeland, while the central valley transitions into extensive cropland and pasture. Forested wetlands follow the Salinas River through the agricultural corridor, and urban development is concentrated around the city of Salinas.

Cropland

For cropland, riparian forest buffers, nutrient management (level 2), conservation tillage (high residue / >60%), and cover cropping were evaluated across 25% of existing acreage. To reduce the high BOD and N/P loads, riparian forest buffers will filter organic matter and dissolved or particulate nutrients. Improved nutrient management and conservation tillage will help lessen N and organic matter loads on cropland that heavily contribute to runoff pollutants. Cover crops will improve nutrient uptake and sediment retention by building up soils.

Pastureland

For pastureland, riparian grass buffers, livestock exclusion fencing, rotational grazing management, and streambank protection were evaluated across 25% of existing acreage. Riparian grass buffer strips help filter and prevent manure, organic solids, particulate P, and N from directly entering waterways. Livestock exclusion fencing prevents direct livestock access and manure pollution in waterways. Rotational grazing helps prevent excessive sediment erosion and improves vegetative cover that can help filter excessive runoff nutrients. Finally, streambank stabilization directly targets BOD and sediment pollution.

Urban

For urban areas, bioretention, infiltration devices, extended wet detention ponds, vegetated filter strips, and wetland detention were evaluated across 25% of existing land area as outlined in **Table 6.3.1**. Bioretention had the highest nutrient reductions among feasible BMPs and is well-suited for urban retrofits of planters and berms. Infiltration basins are well suited for areas with pervious spaces such as road edges or parks and demonstrated strong reductions for all major pollutants. Extended wet detention basins are best suited for areas with available open space and offer strong multi-pollutant reduction. Vegetated filter strips or bioswales are low-cost, easily implemented solutions that work well near roads, parking lots and residential developments.

Table 6.3.1: Urban BMPs by pollutant category. Source: U.S. EPA PLET.

Watershed	Commercial	Industrial	Institutional	Transportation	Multi Family	Open Space
180600051509 - Alisal Creek-Salinas River	Bioretention facility	Infiltration Devices	Extended Wet Detention	Vegetated Filter Strips	Bioretention facility	Wetland Detention

Combined BMP results

Modeling using the U.S. EPA's Pollutant Load Estimation Tool (PLET) provided the impact of the BMPs described above when implemented across 25% of the existing land area in the Alisal Creek - Salinas River watershed area (HUC 180600051509) that was the largest contributor to pollutant loads. Together, these cropland and pastureland BMPs drove 4-21% reductions in pollutant loads with the most substantial reductions in sediment (21%) and Phosphorus (18%) as shown in **Table 6.3.1** and visualized in **Figure 6.3.2**.

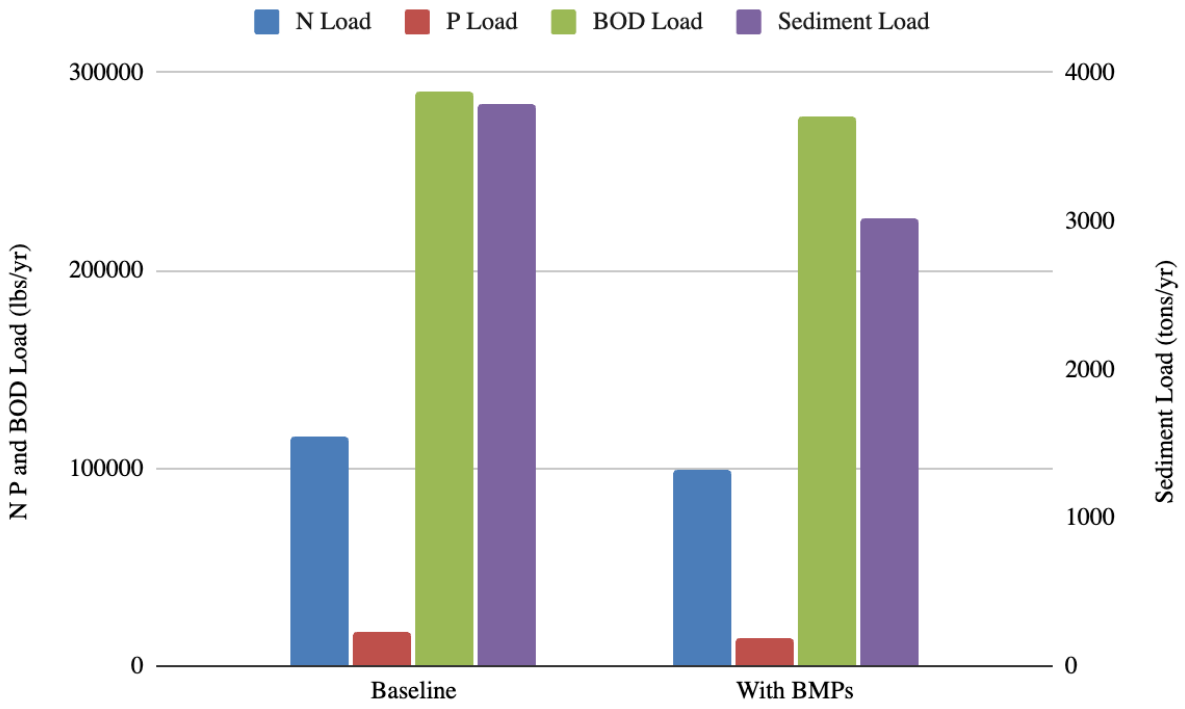

Figure 6.3.2: Comparison between baseline pollutant loads (left) and pollutant loads after BMP application across 25% of acreage (right) in the HUC 180600051509 (Alisal Creek-Salinas River) area most responsible for pollutants. Source: U.S. EPA PLET.

Table 6.3.1.: Relative improvement in pollutant loads with BMPs applied across 25% of acreage in the HUC 180600051509 (Alisal Creek-Salinas River) area most responsible for pollutants. Source: U.S. EPA PLET.

Impact of BMPs	% Change
% Sediment Reduction	21%
% P Reduction	18%
% N Reduction	15%
% BOD Reduction	4%

6.4 Existing or Planned Water Management Projects

A review of publicly available planning documents, agency reports, and regional water management programs reveals a wide range of ongoing and proposed initiatives across the Salinas River Watershed. The purpose of identifying these efforts is to ensure that this watershed management plan aligns with existing local, regional, and regulatory priorities,

avoids duplication, and supports coherent coordination with agencies and community partners already active in the region. Understanding the scope, goals, and timelines of related programs helps ensure that new recommendations complement established initiatives and allows planners to anticipate stakeholder expectations when presenting proposed strategies.

Several long-standing regional programs form the backbone of water management in the watershed. The Salinas Valley Water Project coordinates reservoir releases from Lakes Nacimiento and San Antonio to stabilize groundwater levels and slow seawater intrusion in the lower basin. These operations support downstream agricultural users, help maintain hydrologic balance during periods of variable rainfall, and remain one of the region's primary tools for managing groundwater supply. Complementing this, the Salinas River Stream Maintenance Program conducts sediment removal, vegetation management, and invasive species control within designated reaches of the river. These actions preserve channel capacity, reduce flood risk to adjacent farmland and infrastructure, and provide a coordinated mechanism for landowners and regulatory agencies to maintain river function. Groundwater management responsibilities are further supported by the Salinas Valley Basin Groundwater Sustainability Agency, which oversees SGMA compliance, maintains groundwater monitoring networks, evaluates seawater intrusion risks, and explores long-term strategies to promote basin sustainability.

In addition to these foundational programs, the watershed is home to several major floodplain restoration and multi-benefit climate resilience projects. The Castroville to the Coast Project proposes restoring a three-mile segment of the Tembladero Slough by setting back agricultural land to widen the floodplain. The project aims to reduce flood hazards, enhance wetland habitat, improve surface water quality, and expand recreational and coastal access opportunities for disadvantaged communities. Similarly, the Old Salinas River (OSR) Project seeks to acquire frequently inundated, low-lying lands to reestablish natural floodplain processes that mitigate flooding and improve habitat quality. The Espinosa Lake Project will increase flood storage across a 10-acre area and enable Monterey County Water Resources Agency (MCWRA) to retain approximately 630 acre-feet of freshwater for later reuse while routing flows through a sinuous treatment wetland designed to improve water quality. Coastal protection efforts are supported by the Salinas River State Beach Dune Restoration and Management Plan (2021), which focuses on removing non-native species and planting native vegetation to create a resilient natural buffer against storm surge, coastal flooding, and sea-level rise. Many of these multi-benefit projects are facilitated through state programs such as the Multi-Benefit Land Repurposing Program (MLRP), which provides funding to transition flood-prone agricultural parcels into conservation and habitat uses.

Additional initiatives focus on water supply reliability, treatment improvements, and regulatory water quality compliance. The City of Gonzales is preparing for growing water demands through new well siting analyses, upgrades within its 5-year Capital Improvement Program, and potential expansions of storage capacity. A long-term objective is advancing recycled water production at the Gonzales Wastewater Treatment Plant to support industrial users and reduce potable water demand. Regionally, the Pure Water Monterey (PWM) program continues to expand its advanced treated recycled water facilities to enhance groundwater recharge, increase drought reserves, and augment agricultural irrigation supplies. Water quality improvements are

further driven by compliance with the Central Coast Agricultural Waste Discharge Order (Ag Order 4.0), which requires growers to adopt practices that reduce nitrate leaching into groundwater. Broader coordination is supported by the Greater Monterey County Integrated Regional Water Management Plan (IRWMP), which aligns partner agencies around shared goals for water reliability, flood management, ecological enhancement, and climate resilience. Local ordinances and public outreach campaigns additionally promote water conservation, pollution prevention, and sustainable landscaping in new and existing developments.

Collectively, these initiatives demonstrate a robust and multi-layered regional effort to address water supply reliability, flood protection, ecological restoration, and water quality challenges across the watershed. This context provides a strong foundation for developing a watershed management plan that is complementary, strategic, and responsive to existing programs while helping to fill remaining gaps in regional water management.

Section 7: Recommendations

7.1 Introduction

Sustainable water management in the Salinas River Watershed requires a coordinated set of actions that address groundwater overdraft, seawater intrusion, declining water quality, and variability in surface-water availability. The analyses in the preceding sections highlight the interconnected nature of these challenges: groundwater pumping continues to exceed natural recharge, winter storm flows are highly variable, and agricultural and urban demands remain the dominant drivers of long-term water use. At the same time, the watershed's hydrologic systems provide opportunities for managed recharge, recycled-water expansion, stormwater capture, and supplemental supply development.

This chapter outlines a comprehensive set of recommendations to support basin sustainability through 2045. The recommended actions focus on improving water-supply reliability, enhancing recharge capacity, reducing pressures on overdrafted aquifers, and expanding the use of alternative water sources. Together, these strategies provide a framework to help the region meet Sustainable Groundwater Management Act (SGMA) requirements while maintaining the agricultural productivity and urban water needs that define the Salinas Valley. The following sections present specific recommendations for water-supply diversification, agricultural groundwater management, and implementation of best management practices across major land-use categories.

7.2 Water Supply Recommendation

Ensuring a reliable water supply for the Salinas River Watershed will require strategies that reduce dependence on overdrafted aquifers, expand the use of non-potable alternatives, and capture stormwater more effectively during wet periods. The following recommendations build on the watershed's hydrologic patterns, projected urban and agricultural demands, and the management priorities identified throughout this report.

1. Groundwater Recharge Enhancement

Increasing groundwater recharge in high-infiltration areas is essential for stabilizing conditions in the Eastside and 180/400-Foot Aquifer Subbasins. Alluvial zones along the Salinas River and Arroyo Seco already function as primary recharge corridors and should be prioritized for managed aquifer recharge. Appropriate strategies include construction of off-channel recharge basins, diversion of high winter flows into infiltration ponds during storm events, and temporary use of fallowed agricultural fields for seasonal spreading. Locating recharge projects near the seawater-intrusion front can help maintain freshwater gradients and protect municipal wellfields. Expanded monitoring of groundwater levels, salinity, and infiltration performance is necessary to evaluate long-term effectiveness.

2. Recycled Water Expansion

Recycled water offers a drought-resilient supply that can reduce potable-groundwater demand in both urban and industrial sectors. Gonzales and Cal Water have already identified tertiary recycled water as a future supply source, and expansion of reuse should begin with landscape irrigation, industrial cooling, and commercial applications. Over time, upgraded treatment and distribution systems could support indirect potable reuse by delivering highly treated recycled water to recharge basins in the Eastside Subbasin. This approach would strengthen municipal supply reliability while contributing to seawater-intrusion control. Successful implementation will require treatment upgrades, new conveyance infrastructure, and clear public communication to ensure confidence in water-quality safeguards.

3. Stormwater Capture and Infiltration

Urban stormwater represents an underutilized opportunity to enhance local recharge, particularly in Salinas, Gonzales, and Soledad where impervious surfaces limit infiltration. Integrating Low Impact Development features into new Specific Plan areas and redevelopment projects—such as bioretention basins, vegetated swales, permeable pavement, and rain gardens—will increase stormwater retention and improve downstream water quality. Larger neighborhood-scale detention and infiltration systems should be incorporated into planned growth areas to attenuate peak flows and contribute to groundwater recharge during winter storm events. Where feasible, stormwater infrastructure should provide multiple benefits by combining infiltration, open space, and habitat improvement within the same project footprint.

4. Desalination as a Supplemental Supply

Desalination should be evaluated as a long-term supplemental supply to reduce reliance on overdrafted coastal aquifers. Existing facilities at Moss Landing and Sand City demonstrate regional feasibility and present opportunities for cost-effective partnerships that would allow Salinas Valley communities to secure a contracted supply volume without constructing new facilities. A small-scale coastal desalination plant designed to support indirect potable reuse or groundwater recharge could also be considered to help maintain freshwater gradients inland of the seawater-intrusion front. Any desalination expansion must comply with California Ocean

Plan requirements, address brine-discharge and intake impacts, and incorporate renewable-energy procurement to mitigate energy demands. Desalination should complement, not replace, conservation, expansion of recycled water, or stormwater capture.

7.3 Groundwater Recommendations for Agriculture

While cropland irrigation in the Salinas Valley is currently operating efficiently—the 0.94 ft/acre/yr is in line or below statewide estimates—the sheer amount of cropland that requires irrigation and the depleted groundwater storage still necessitates further refinement. Upgrading aging irrigation systems offers one of the most direct and impactful ways to reduce groundwater extraction.

1. Grower-led measurement and outreach plan

In the same way that the wide dispersal and high variability of agricultural and ranching practices in the Salinas Valley have led to monitoring gaps for non-point sources as detailed throughout the management plan, there are also information gaps between growers and ranchers. These gaps present an opportunity for improved coordination and knowledge-sharing to ensure that the most water efficient technologies are being used as widely as possible. However, this cannot be achieved through traditional top-down distribution tactics and new technologies often require proof of concept within the specific growing context of a region before adoption will spread.

Building on the successful enrollment and participation in the grower-led Cooperative Monitoring Program which established 50 WQ monitoring sites, the coalition should continue the momentum to establish water quality and efficiency improvements as the natural next step for land managers to take full ownership over the long term viability of their water supply. As more non-point source specific data becomes available, growers will be empowered to make better-informed decisions about their land management practices and Central Coast Water Quality Preservation, Inc. can step back in to help offer guidance with a recommended toolkit of BMPs such as irrigation efficiency and precision agriculture to meet their needs while minimizing excessive water use and pollutant loads. As these practices are established by first-movers in the Valley and proven through workshops and on-site demonstrations by CCWQP to improve efficiency while reducing operating costs, adoption rates will hopefully continue to accelerate.

2. Establish an Irrigation Efficiency Certification

Although much of the Salinas Valley has already transitioned to drip irrigation for high-value crops, many systems may be inefficient due to uneven pressure, clogged emitters, line breaks and poorly calibrated flow rates. A coordinated initiative to modernize irrigation could address these inadequacies. Growers who meet the following criteria and fall within efficiency benchmarks are eligible for reduced pumping fees. Requirement of recertification every three years to disincentive increased pumping after receiving reduced fees.

- Replace older drip lines with pressure-compensating emitters that maintain uniform distribution across longer beds and uneven topography
- Convert any remaining sprinkler systems to micro-sprinklers or duel drip systems
- Install automated pressure and flow sensors to detect leaks, failures, or pressure drops

All certified growers are entered into a quarterly raffle for farming equipment and irrigation upgrades. With the efficiency upgrades from the certification program, water application rates could decrease while improving yield consistency.

3. **Deploy Precision Agriculture for Real Time Irrigation Inputs**

Leveraging real-time monitoring technologies—combined with machine learning—can significantly optimize irrigation timing and quantity.

- Remote sensing and drone-based irrigation monitoring. Multispectral drone imagery can detect early water stress through canopy temperature, NDVI, and spectral signatures like Solar Induced Chlorophyll Fluorescence which is invisible to the human eye. Thermal imaging identifies uneven irrigation patterns, clogged emitters or areas receiving more water than needed.
- Machine Learning models for irrigation scheduling. Models trained on soil moisture, ET data, crop type, weather forecasts and canopy signals can recommend real-time irrigation volumes.
- Soil Moisture and tensiometer networks. Distributed sensors allow fields to be irrigated only when the root zone approaches a crop-specific threshold
- Data streams can feed into cloud-based dashboards that aggregate soil moisture trends across fields

7.4 Recommended BMPs

The initial BMP analysis conducted using the EPA's PLET assigned four BMP's to each major land use category and measured overall changes to observed pollutant loads as outlined in Section 6.3. The top three BMPs are highlighted here with details on site selection and direct actions for the most successful implementation.

Cropland BMP Recommendations

1. **Riparian forest buffers along the Salinas River and tributaries**
    - Prioritize reaches near Gonzales and Soledad where cropland extends to channel edges.
    - Provides filtration for N, P, and BOD; reduces channel erosion as evidenced in **Figure 7.4.1**.

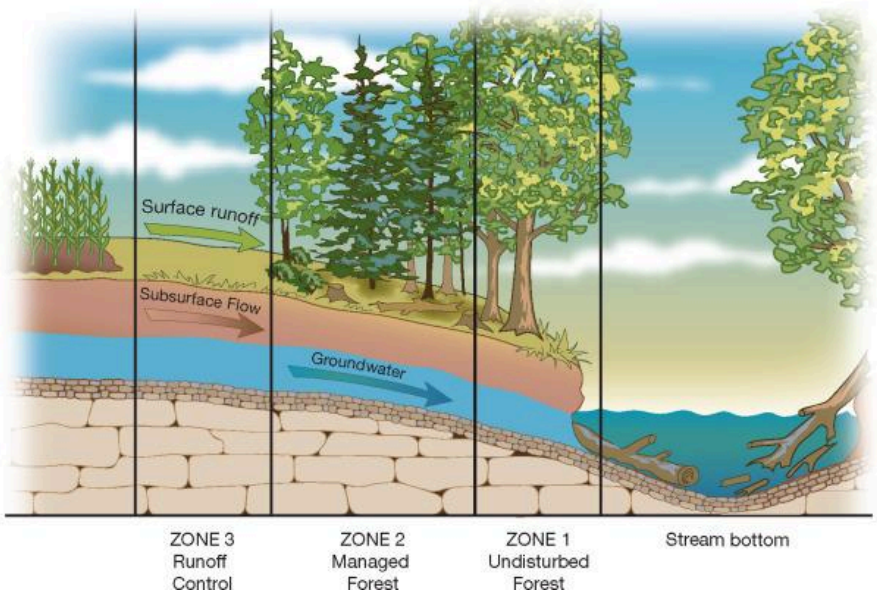

Figure 7.4.1: Diagram of Riparian Forest Buffer. Source: Lowrance et al. 1995

2. **Expanded Conservation Tillage in the southern agricultural corridor**
    - Target fields with high erodibility.
    - Reduces sediment loads during flashy storm pulses.
3. **Cover Cropping demonstration network**
    - Start with three pilot blocks (one per city boundary).
    - Demonstrate winter nutrient uptake benefits and soil organic matter increases.
    - Partner with UC Cooperative Extension programs and research center, specifically Timothy Bowles' lab at UC Berkeley.

Pastureland BMP Recommendations

1. **Livestock Exclusion fencing in high-sediment pasture ranches**
    - Especially effective in upper tributaries feeding Alisal Creek-Salinas River HUC.
    - Reduces direct manure inputs and prevents bank collapse.
    - Host fence maintenance and construction workshops to provide ranchers with adequate resources and knowledge base.
2. **Rotational Grazing partnerships with local producers**
    - Pilot with large ranches in the foothills of Gabilan and Santa Lucia Ranges.
    - Aligns forage rotation with soil moisture and storm timing to limit erosion.
    - Utilize satellite data to track progress.
3. **Streambank Stabilization using bioengineering**
    - Install willow, mule-fat, and sedge revetments to reinforce eroding banks.
    - Align with Stream Maintenance Program to maximize efficiency.

Urban BMP Recommendations

1. **Green Infrastructure Corridors in Salinas**
    - Build bioretention planters along South Main St. and East Alisal Rd.

    - Integrate with planned complete-streets upgrades in WASP and CASP.
2. **Infiltration Devices in Soledad and Gonzales**
    - Use parks, schoolyards, and open spaces as infiltration hubs.
    - Addresses urban contribution to nutrient loads.
3. **Extended Wet Detention Basins in New Developments**
    - Pair with Gonzales' new Community Commercial Mixed-Use area.
    - Effective for capturing moderate (2.5–5 cm) storm events shown in the storm distribution.

Conclusion

The Salinas River Watershed faces interconnected challenges of groundwater overdraft, water quality degradation, and variable water availability that require coordinated management through 2045. This plan provides a comprehensive framework addressing these challenges through groundwater recharge enhancement, recycled water expansion, stormwater capture, and targeted best management practices that align with existing regional initiatives and SGMA requirements. By implementing the recommended strategies for water supply diversification, agricultural efficiency, and pollution control across cropland, pastureland, and urban areas, the region can maintain its position as the nation's agricultural leader while ensuring long-term ecological health and water sustainability for all beneficial uses.

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Appendix

Table A1: Designated and Beneficial Uses for Water Bodies in Salinas River Watershed. Beneficial uses are organized by water body type and include existing uses (groundwater for municipal and agricultural supply, riparian habitat, recreation), protection goals (groundwater recharge, flood control, critical habitat preservation), and potential future uses (recycled water for irrigation and recharge). Use designations that follow California Regional Water Quality Control Board Basin Plan nomenclature and represent the basis for establishing water quality criteria and management priorities. Information was synthesized from General Plan Documents and experts from the Groundwater Sustainability Plan (GSP).

Water Body / Water Type	Beneficial Use Category	Specific Use

Groundwater (Salinas Valley Basin, 180/400-Foot Aquifer)	Municipal and Domestic Supply (MUN)	Existing Use Sole Source of Domestic Water
	Agricultural Supply (AGR)	Existing Use
	Industrial Supply (IND/PROC)	Existing Use
	Groundwater Recharge (GWR)	Potential Use
Salinas River (Segment near Gonzales WWTP)	Rare, Threatened, or Endangered Species (RARE)	Existing Critical Habitat
	Wildlife Habitat (WILD)	Existing Habitat
	Groundwater Recharge (GWR)	Protection Goal
	Flood Flow Capacity / Control	Protection Goal
Gonzales Slough and Drainages (Johnson Canyon Creek, McCoy Creek)	Passive Recreation (REC-2)	Existing/Planned Use
	Wildlife Habitat (WILD)	Existing Habitat
	Drainage and Flood Control	Existing Use Public Health & Safety
	Agricultural Drainage	Existing Use
Inland Surface Waters (General, Central Coast Basin)	Recreational Water (REC-1/REC-2)	Present/Potential Use
	Warm Freshwater Habitat (WARM)	Present/Potential Use
	Spawning, Reproduction, and/or Early Development (SPWN)	Present/Potential Use
Recycled Water (Future/Potential)	Agricultural Supply (AGR)	Potential Future Use
	Industrial Use (IND/PROC)	Potential Future Use
	Landscape Irrigation	Potential Future Use

	Groundwater Recharge (GWR/IPR)	Potential Future Use
	Habitat Enhancement / Wetlands (WILD/BIOL)	Potential Future Use
Ocean/Coastal Waters (General Monterey Subbasin)	Preservation of Biological Habitats of Special Significance (BIOL)	Protection Goal

Table A2. Log of Contaminants at Various Monitoring Locations

ID	Location	Latitude	Longitude	Parameter	Count of Value	Max of Value	Units
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	ACETAMIPRID	2	0.0105	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	ALLETHRIN	1	-0.00005	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	CLOTHIANIDIN	1	0.348	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	CYFLUTHRIN	1	0.00287	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	DINOTEFURAN	1	-0.0006	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	FENPROPATHRIN	1	-0.00003	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	FENVALERATE	1	-0.00005	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	FLUVALINATE	1	-0.00005	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	PRALLETHRIN	1	-0.00005	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	THIACLOPRID	1	-0.0002	ug/L
CEDEN-309 HRT	Alisal Creek @ Hartnell Rd dogleg	36.6407	-121.579	THIAMETHOXAM	2	0.77	ug/L
CEDEN-309-ASILO-31	Asilomar State Park at bridge	36.6171	-121.939	Copper	744	600	ug/L
CEDEN-Carm Riv1	Carmel River Beach mussels	36.539	-121.932	BIFENTHRIN	5	0.0402	ug/L
CEDEN-Carm Riv1	Carmel River Beach mussels	36.539	-121.932	FIPRONIL SULFONE	6	0.00159	ug/L
CEDEN-307-CASD-02	Carmel Storm Drain at the end of Ocean Avenue	36.555	-121.93	MBAS	76	1.84	mg/L
CEDEN-306-CARNE-36	Carneros Creek at Blohm Road	36.86	-121.74	Suspended Sediment	58	691.2	mg/L

				Concentration (SSC)			
CEDEN-306-CARNE-36	Carneros Creek at Blohm Road	36.86	-121.74	Kjeldahl nitrogen	46	29	mg/L
CEDEN-306-CARNE-36	Carneros Creek at Blohm Road	36.86	-121.74	.alpha.-Endosulfan	16	106	ug/L
CEDEN-306-CARNE-36	Carneros Creek at Blohm Road	36.86	-121.74	Endrin	15	126	ug/L
CEDEN-306-CARNE-31	Carneros Creek at Johnson Road	36.8606	-121.708	Nitrite	562	13.491	mg/L
CEDEN-306-CARNE-31	Carneros Creek at Johnson Road	36.8606	-121.708	Organic carbon	129	45.55	mg/L
CEDEN-306-CARNE-31	Carneros Creek at Johnson Road	36.8606	-121.708	Fluoride	75	10.7094	mg/L
CEDEN-306-CARNE-31	Carneros Creek at Johnson Road	36.8606	-121.708	Bromide	51	58.9512	mg/L
CEDEN-306-CARNE-31	Carneros Creek at Johnson Road	36.8606	-121.708	.beta.-Endosulfan	15	88	ug/L
CEDEN-306-CARNE-31	Carneros Creek at Johnson Road	36.8606	-121.708	p,p'-DDT	14	81.9	ug/L
CEDEN-306-CARNE-32	Carneros Creek at mid entry to wetland	36.8614	-121.717	Phosphorus	108	14	mg/L
CEDEN-306-CARNE-32	Carneros Creek at mid entry to wetland	36.8614	-121.717	p,p'-DDE	15	75.5	ug/L
CEDEN-306-CARNE-32	Carneros Creek at mid entry to wetland	36.8614	-121.717	Heptachlor epoxide	15	164	ug/L
CEDEN-306-CARNE-32	Carneros Creek at mid entry to wetland	36.8614	-121.717	p,p'-DDD	14	85.5	ug/L
CEDEN-306-CARNE-32	Carneros Creek at mid entry to wetland	36.8614	-121.717	Malathion	13	0.25	ug/L
CEDEN-306-CARNE-33A	Carneros Creek- Sill Road, road flow	36.8618	-121.723	Nitrogen	54	17.656	mg/L
CEDEN-306-CARNE-33A	Carneros Creek- Sill Road, road flow	36.8618	-121.723	Dieldrin	16	115	ug/L
CEDEN-306-CARNE-33A	Carneros Creek- Sill Road, road flow	36.8618	-121.723	Lindane	14	69	ug/L
CEDEN-309-CENTR-31	Central and 13th Streets in Pacific Grove in Greenwood Park	36.6216	-121.914	Total suspended solids	825	10460	mg/L
CEDEN-309-CENTR-31	Central and 13th Streets in Pacific Grove in Greenwood Park	36.6216	-121.914	Sulfate	81	185	mg/L
CEDEN-309SOS	Chualar Creek @ Old Stage Rd (south branch)	36.5456	-121.47	Nitrate + Nitrite	603	210	#N/A
CEDEN-306ELKERD	Elkhorn Slough Upper near RR tracks	36.8566	-121.758	Alkalinity, total	37	345	mg/L
CALWR_WQX-D4361020	LITTLE SUR R A HWY 1	36.3316	-121.8861	Specific conductance	91	2827	uS/cm

CEDEN-309-MAJOR-31	Majors Creek lower	36.5925	-121.892	Lead	429	283	ug/L
CEDEN-309 MTRY24	Monterey Harbor random site 24	36.6072	-121.891	Salinity	52	37.2	ppt
CEDEN-309 MTRY28	Monterey Harbor random site 28	36.6036	-121.891	Dissolved oxygen (DO)	156	10.16	mg/L
CEDEN-309 MTRY28	Monterey Harbor random site 28	36.6036	-121.891	Survival	123	100	%
CEDEN-309 MTRY08	Monterey Harbor random site 8	36.602	-121.89	Ammonia	619	64.9	mg/L
CEDEN-309-MSD-04	Monterey storm drain at San Carlos Beach	36.6095	-121.895	Nitrate	1376	314	mg/L
CEDEN-309-MSD-04	Monterey storm drain at San Carlos Beach	36.6095	-121.895	Zinc	705	1170	ug/L
CEDEN-309-MSD-04	Monterey storm drain at San Carlos Beach	36.6095	-121.895	Magnesium	612	166	mg/L
CEDEN-309-MSD-05	Monterey storm drain at Steinbeck Plaza	36.6161	-121.9	Turbidity	125	1800	NTU
CEDEN-309-MSD-05	Monterey storm drain at Steinbeck Plaza	36.6161	-121.9	Chloride	82	268	mg/L
CEDEN-306-MC-SAL-12	Moro Cojo Salinity Transect 12	36.7778	-121.738	Ammonium	407	51	mg/L
CEDEN-306 MSLG14	Moss Landing Harbor random site 14	36.8063	-121.785	Silt	60	54.9	%
CEDEN-309 MSLG30	Moss Landing Harbor random site 30	36.8031	-121.786	Sand	76	78.1	%
CEDEN-309 MSLG04	Moss Landing Harbor random site 4	36.8026	-121.786	Moisture content	136	66.81	%
CEDEN-309 MSLG04	Moss Landing Harbor random site 4	36.8026	-121.786	Hydrogen sulfide	19	0.2252	mg/L
CEDEN-306 MSLG06	Moss Landing Harbor random site 6	36.8116	-121.788	Chlorophyll a	149	452.3205	ug/L
CEDEN-306SAS	Old Salinas River @ Sandholdt Bridge	36.8003	-121.787	IMIDACLOPRID	5	0.399	ug/L
CEDEN-309-PGSD-09	Pacific Grove storm drain at HopkinsMon	36.619	-121.903	Urea	324	5210	ug/L
CEDEN-309-PGSD-03	Pacific Grove storm drain at Lovers Point Beach	36.6247	-121.918	Total dissolved solids	183	3590	mg/L
CEDEN-309-PGSD-03	Pacific Grove storm drain at Lovers Point Beach	36.6247	-121.918	Conductivity	135	25260	uS/cm
CEDEN-309-PGSD-03	Pacific Grove storm drain at Lovers Point Beach	36.6247	-121.918	Potassium	104	23	mg/L
CEDEN-309-PGSD-04	Pacific Grove storm drain between Pico and Arena Avenues	36.6249	-121.94	Calcium	681	347	mg/L

CEDEN-309-PGSD-04	Pacific Grove storm drain between Pico and Arena Avenues	36.6249	-121.94	Iron	14	3060	ug/L
CEDEN-305PJE00L	Pajaro River Estuary Lower, below confluence with Watsonville Slough	36.8516	-121.808	FIPRONIL DESULFINYL	6	0.0051	ug/L
CALWR_WQX-D2116020	SALINAS R A DAVIS RD	36.6416	-121.7011	pH	238	78	None
CALWR_WQX-D2116020	SALINAS R A DAVIS RD	36.6416	-121.7011	Temperature, water	81	27	deg C
CEDEN-309SAL00U	Salinas River Estuary Upper near RR bridge	36.7314	-121.782	ALUMINUM	2	59.7	ug/L
CEDEN-SalRiv2	Salinas River POP	36.695	-121.75	FIPRONIL	4	0.0056	ug/L
CEDEN-SalRiv2	Salinas River POP	36.695	-121.75	FIPRONIL SULFIDE	6	0.00087	ug/L
CALWR_WQX-D3209830	SAN ANTONIO R BL SAN ANTONIO DM	35.8036	-120.8555	Oxygen	21	88.3	mg/L